

2013

Annual Water Report

Prepared by:

Town of Ladysmith

Infrastructure Services

Department

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1. Introduction

The 2013 Annual Water Report provides an overview of the Town of Ladysmith water system (water sources, maintenance programs & capital improvements) and summarizes the annual water quality and production data. All water suppliers, under their Operating Permit, are required to provide an annual water report to the Vancouver Island Health Authority. This report is also posted on the municipal web site at www.ladysmith.ca.

2. Service Area & Sources

The Town of Ladysmith's water is drawn from two sources, which provide water through separate facilities at the mid/north and south ends of Town. The community is supplied in part by the Holland Lake and Holland Creek watershed which enters the water supply system at a diversion point at the Chicken Ladder Intake, a small stilling basin located approximately 2.5 km. from the Public Works yard, as well as from Stocking Lake, located south of the Town. Both of these supplies are now piped directly to the enclosed Arbutus Reservoir, where it is chlorinated and distributed by gravity to the entire Ladysmith service area and the Diamond Improvement District.

Holland Lake

The Holland Lake reservoir was constructed in 1979 at the location of two original lakes approximately 5 km west of the Town in a 'regulated access' watershed – it consists of two earth filled embankments, providing 1,600,000 cm of live storage. The reservoir currently discharges through a 450mm dia. steel pipe located 6.0 meters (19.7 feet) below the spillway elevation. There is a small remotely operated valve that controls discharge into Holland Creek, which augments base flows in Holland Creek, and enters the Town's water system at Chicken ladder, located approximately 1 km west of the Arbutus reservoir. The dam is in good condition, and receives regular maintenance in accordance with current Provincial Dam Safety Standards. The dam is rated a high risk structure in accordance with the Provincial Dam Classification system. In late 2012, a remote turbidity meter was installed at the dam to measure turbidity year round. Minor repairs were also undertaken in 2013 to improve emergency spillway operation at the west embankment.

The Holland Lake reservoir is located predominately in active forestry lands owned by TimberWest Forest Corporation (TimberWest). The majority of lands contributing to the lower reaches of Holland Creek upstream of Chicken Ladder intake is crown land managed forest. The Town owns the Holland Lake perimeter, while the lands contributory to the Lake is owned by TimberWest.

Logging Roads exist throughout the area, but are gated and signed to restrict public access to the Town's Lakes and Intakes. The public use the watershed for walking, and other related recreational activities, however, camping and water use at Holland and Stocking Lakes is restricted.

Chicken Ladder Intake (Holland Creek)

The Chicken ladder Intake was constructed on Holland Creek in the 1960's, when the lower Holland Dam water intake that was originally constructed by the Ladysmith Water Company was relocated due to the need for higher operating pressures in the system. At that time, an earthen open reservoir was constructed adjacent to Holland Creek under the Power Lines, and a chlorination building was constructed at the same time. The earthen open reservoir was subsequently filled in and replaced with a 5,700 cm concrete covered reservoir in 2008, and the

replacement of the chlorination building is under way in 2013/2014, which will be located on the site of the old open reservoir. The Chicken Ladder intake is located next to a well-used trail, and is considered to represent a moderately high risk for public access due to poor site security and constrained site access due to topography. The site is signed and marked restricting public access, but the remoteness of the location results in less than ideal site security.

Stocking Lake

Stocking Lake was built in the 1920's, by the Ladysmith Water Company, which was subsequently taken over by the Town. The lake serves both the Town, as well as the Saltair area (Cowichan Regional District). The Intake, which is shared by both water jurisdictions, is a 900mm dia. steel pipe located 5.0 meters (16.4 feet) below the spillway elevation. The dam is in 'fair' general condition, but has a documented seepage issue that dates back to the 1980's. The amount of seepage does not appear to change from year to year, however, discussions have commenced between the CVRD and the Town to repair the leak by adding additional embankment and enhance the stability of the structure. The dam is rated a low risk structure in accordance with the Provincial Dam Classification system.

The Stocking Lake area also is frequented by ATV users and occasional campers, as there is evidence of vandalism in the area. There is a trail system that travels the length of the lake. The contributing areas of the watershed are owned by TimberWest, the Crown, the CVRD, and the Town.

Arbutus Reservoir

The Arbutus reservoir, 5,700 cu meters in capacity, currently provides daily peaking for the Town. As of the fall of 2012, both the North as well as the South portions of Ladysmith are served from the Arbutus Reservoir.

Figure one shows the general layout of the Watersheds and major storage and intake structures.

2.1.Ladysmith Water System - Distribution

The topography of the Town of Ladysmith slopes steeply towards the waterfront, resulting in a large range in the water pressures within the Town's water system. With the completion of the covered Arbutus Reservoir in 2008 and subsequent increase in distribution pressure (32psi), the Town introduced two specific pressure zones within the system with the installation of four pressure reducing stations. Future development above the 121 (approximate) meter contour will require the construction of additional reservoirs and pumping facilitates.

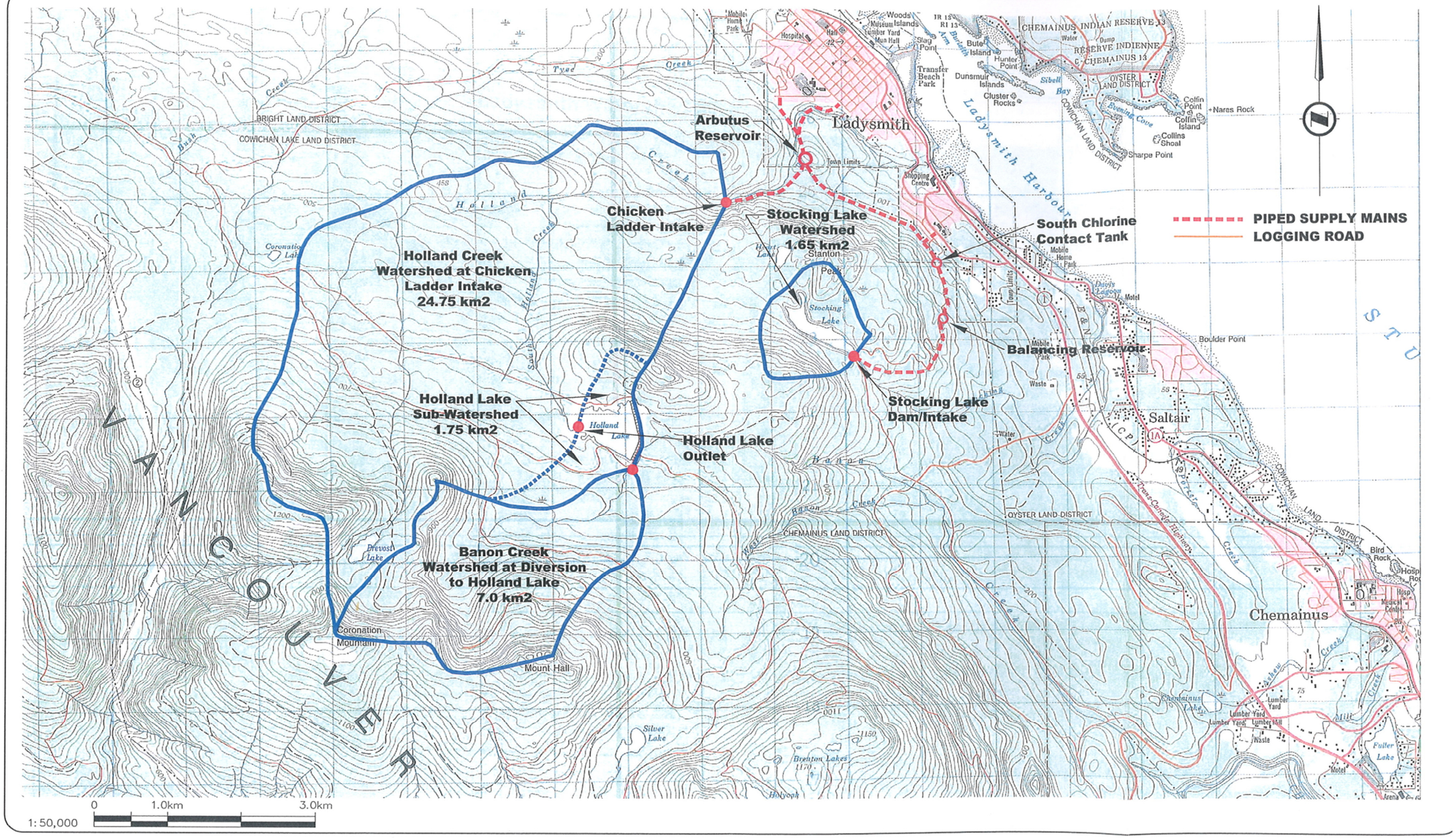
The distribution system for the Town consists of over 61 km of water main of varying sizes from 100mm to 250 mm. The mains are made of cast iron piping, asbestos cement piping or PVC piping. The cast iron and the asbestos cement piping (approximately 21 km) are located mainly in the older section of Town and within the area that was amalgamated from Saltair in 1985. The majority of the asbestos cement piping is 100 mm in diameter and needs to be replaced to allow for adequate flows throughout the system.

The Town has been actively working on a replacement program since 2000. Fire hydrants are situated throughout the distribution system at a spacing to offer adequate fire flows for all properties.

The Town has had access to computer modelling software for monitoring static and dynamic water pressures within the system. Plans are under way to provide this modelling capacity internally to allow integration of modeling software with asset management systems.

The Town is in the process of replacing the existing gas chlorination system with a new gas system, with provision to possibly manufacture chlorine in the future. The project, scheduled for completion in 2014, is part of a multi-phase project to improve water treatment for the Town. This is discussed later in the Report.

Figure 1 System Overview



3. Water Licensing

The Town currently holds the following water licenses:

Table 1 - Town Water Licences

License	Type	Annual Storage (cm, or Diversion (cm/yr)	Max Day Flow (Cu m)	Location	Date of Original Issuance	Comments
CL 017746	Diversion	995,000		Holland Creek (Chicken Ladder)	1946	Whole Year
CL 029821	Storage	123,348		Holland Creek (Chicken Ladder)	1962	Original earth reservoir (Now replaced with Arbutus Reservoir)
CL 125167	Diversion		3,640 cm/ day	Banon Creek	1977	Replaces 112813 Diversion
CL 125167	Storage	1,820,000		Holland Lake	1977	Replaces 112813 Storage, updated to allow Holland to Stocking Pipeline
CL 112812	Storage	246,700		Holland lake	1952	Replaces 21164
CL 005333	Diversion	Either: 829,000, or 454,000*		Stocking Lake	1902	Max 2,273 cm/ day

**Under review by TOL staff*

Of relevance to the Town, the Cowichan Valley Regional District (CVRD) also holds water licenses on Stocking Lake as follows:

Table 2 - CVRD Water Licenses

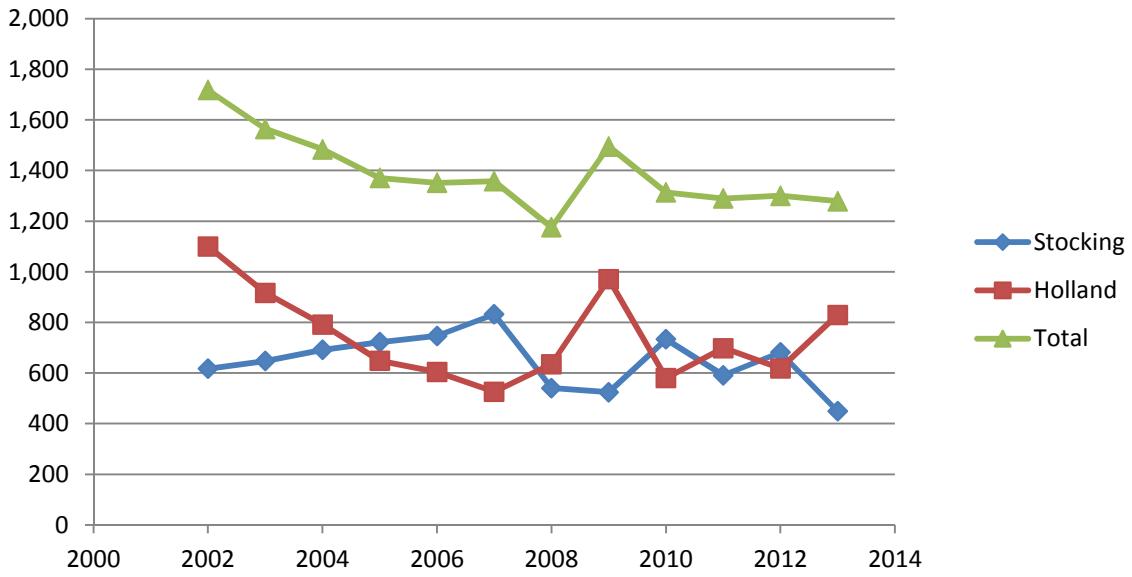
License	Type	Annual Storage (cm, or Diversion (cm/yr)	Max Day Flow (Cu m)	Location	Date of Original Issuance	Comments
CL 67482	Diversion	476,000		Stocking Lake	1984	All year, replaces 61318
CL 67481	Diversion	447,000		Stocking Lake	1955	All year, replaces 28487
Cl 67484	Storage	542,000		Stocking Lake	1984	

4. Water Consumption

The Town used 1,279,000 cubic meters of water in 2013, a reduction of 1.5% over 2012. This reflects a trend that has been occurring at the Town since residential water metering was introduced in 2006. The Town is using approximately 25% less water now than 11 years ago (2002 annual consumption was 1,700,000 cu m). During the same period, the Town has grown over 20%, resulting in a net reduction of per capita water consumption in the order of 45%.

A summary of water consumption by month, including the flow splits between Stocking Lake and Chicken ladder Intake (Holland Creek) follows on the next page, and is illustrated as follows:

Figure 2
Ladysmith Total Consumption and Flow Source – 2002-2013
(In Thousands of cubic meters)



The general downward trend in water consumption is likely the result of the Town's decision to install residential water meters in 2006, as well as its progressive block pricing rate structure for single family residential water accounts. The Town has also been encouraging water conservation through programs such as a toilet rebate program, which would also have a downward impact on consumption.

Flow Split between Holland and Stocking Lakes

Over the last 10 years, the Town has used roughly equal amounts of water from Holland and Stocking watersheds. This has occurred primarily for two reasons: Firstly, until the fall of 2012, portions of South Ladysmith were not able to be fed of the Holland Watershed due to piping

Table 3 Town of Ladysmith Water Consumption, 2002-2013

(In thousands of cubic meters)

Year	Totals			Jan			Feb			Mar			Apr			May			Jun			Jul			Aug			Sep			Oct			Nov			Dec		
	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total	Stocking	Holland	Total
2002	617	1,101	1,718	112	2	114	100	1	102	78	34	112	63	48	110	18	126	144	21	176	198	22	196	218	22	207	230	15	148	163	9	113	122	80	22	102	77	27	104
2003	647	917	1,565	101	7	108	90	15	105	84	24	108	91	11	102	11	121	132	20	186	206	23	194	217	15	155	170	8	103	112	54	47	101	49	52	101	101	2	103
2004	691	792	1,484	112	3	115	103	2	105	103	1	104	36	80	116	13	140	153	14	142	157	21	150	171	14	138	153	25	84	109	53	49	102	97	3	100	98	0	98
2005	722	649	1,371	113	0	113	99	0	99	101	0	101	90	8	98	46	77	123	11	117	128	15	134	149	20	152	173	16	118	134	45	43	88	75	0	75	92	0	92
2006	747	605	1,352	103	0	103	76	1	77	90	0	90	85	0	85	122	11	133	50	84	134	22	162	183	21	155	176	25	108	133	17	84	102	77	0	77	60	0	60
2007	832	526	1,358	109	2	111	98	0	98	102	0	102	99	4	102	66	61	127	22	126	148	28	141	168	25	61	86	17	102	119	87	11	97	73	19	92	106	1	107
2008	541	635	1,176	76	0	76	35	0	35	62	0	62	33	0	33	61	51	112	20	118	139	34	163	197	22	128	150	23	86	109	79	8	86	64	18	82	32	63	95
2009	524	971	1,495	52	43	96	30	56	86	42	48	90	39	80	119	30	134	164	36	160	196	31	158	189	26	126	151	24	82	107	61	35	96	99	2	102	54	45	100
2010	734	580	1,314	94	1	95	67	19	85	54	51	105	58	29	86	54	48	102	37	74	112	26	144	170	23	127	150	43	65	108	85	8	94	95	3	98	98	11	109
2011	591	698	1,290	99	0	99	49	34	84	78	18	96	53	37	90	67	31	98	15	115	130	19	135	154	20	140	160	29	89	118	78	13	91	49	36	84	36	51	87
2012	682	619	1,300	74	15	88	80	0	80	91	0	91	86	0	86	32	80	112	14	103	117	20	137	157	23	142	165	19	105	124	65	36	101	88	0	88	92	0	92
2013	**449**	**830**	**1,279**	**90**	**0**	**90**	**86**	**0**	**86**	**93**	**3**	**96**	**46**	**38**	**85**	**15**	**93**	**108**	**24**	**95**	**118**	**2**	**169**	**171**	**1**	**149**	**150**	**12**	**94**	**106**	**20**	**70**	**90**	**53**	**30**	**83**	**7**	**89**	**97**

and chlorination equipment considerations. In 2012, a two way interconnecting pipe system was installed to connect the Stocking Lake supply line directly to the Arbutus Reservoir. Upon completion of this project, the old chlorination facility servicing South Ladysmith was taken out of service, and all chlorination is now taking place near the Arbutus Reservoir. This has had the effect of allowing an additional 100,000 cubic meters of Holland Creek water to be able to serve South Ladysmith during times when Holland Creek has acceptable turbidity levels.

5. Drinking Water Regulations

In Canada, drinking water is regulated by the Provincial and Local Governments. While the Federal Government, Health Canada, performs research and publishes recommendations for safe drinking water, each Province has the responsibility to regulate drinking water (in British Columbia [BC], the Ministry of Health has this responsibility). Locally, each water supplier also complies with the local Health Authority requirements (The Town of Ladysmith is under the jurisdiction of Island Health [VIHA]).

This section presents the Federal recommendations, as well as the BC regulations and requirements from the VIHA.

5.1. Canadian Drinking Water Quality Guidelines

The Guidelines for Canadian Drinking Water Quality are established by the Federal-Provincial-Territorial Committee on Drinking Water and are published by Health Canada (Ref. 8). They are regularly revised, based on the latest research results. The Guidelines are intended to be used as benchmarks for the Provinces.

The Guidelines consider more than 100 parameters that can, potentially, be found in Canadian drinking water sources: bacteriological pathogens; physical and chemical contaminants, including metals, inorganics, pesticides, and other organics; as well as radionuclides. The Guidelines establish two types of limits for these contaminants. The Maximum Acceptable Concentration (MAC) is based on health considerations, while the Aesthetic Objective (AO) is based on aesthetic considerations.

Appendix A provides the latest revision (June 2012) of the Canadian Drinking Water Guidelines.

5.2. BC Regulations

In BC, drinking water is regulated by the British Columbia Drinking Water Protection Regulation (2003) and the Drinking Water Protection Act (2001) (Refs. 1 and 2, respectively).

The BC Ministry of Health is primarily concerned with bacteria. Escherichia coli and fecal coliforms must not be detected. Total coliforms must not be detected 90 percent of the time, and if detected, they must be less than 10 counts per 100 mL.

The regulations also provide monitoring frequency requirements. They vary with the number of served population. For a water supply system that serves between 5,000 and 90,000 people, it is required that 1 sample per 1,000 people be collected per month. For Ladysmith, which currently serves 8,000 people, 8 samples per month are required. Moreover, the bacteriological analyses must be performed by a laboratory which has been approved by the BC Ministry of Health.

Other requirements include reporting. The water supplier must make public an annual report showing the results of the monitoring. Also, if the standards are not met, the laboratory must immediately inform the health officer and the water supplier. The water supplier must then give a public notice of non-potable water.

The regulations also require certification for water systems operators. This will be discussed in section 8.0 of the report.

5.3. Island Health Authority Requirements

The Town must also comply with the local Health Authority requirements. VIHA officers evaluate and assess new sources of water for public use, make recommendations for operating permits, review water quality monitoring data, and inspect water systems.

VIHA has issued two policies regarding drinking water quality. The *Guidelines for the Approval of Water Supply Systems* was issued in 2006 and provides treatment requirements and recommendations on water quality testing before the approval of any new water supply system.

The *Drinking Water Treatment for Surface Water Supplies Policy*, or *4-3-2-1 Policy*, was issued at the end of 2007 and refers to treatment requirements for water systems supplied by surface water. Both are discussed below.

Guidelines for the Approval of Water Supply Systems

VIHA *Guidelines for the Approval of Water Supply Systems* require that before the submission for approval of new water supply system, raw water be characterized for the following parameters:

- Microbiological pathogens: total coliforms, non-coliform bacteria, Escherichia coli,
- heterotrophic plate count
- Physical parameters: colour, conductivity, pH, turbidity
- Chemical parameters: alkalinity, corrosiveness, hardness, organic nitrogen, total
- dissolved solids (TDS), total organic carbon (TOC), ammonia, chloride, fluoride, nitrate,
- nitrite, sulphate, arsenic, selenium, and other metals
- The guidelines also require treatment providing the following levels:
- 2 log inactivation or reduction for Cryptosporidium and 3 log inactivation or reduction for Giardia
- 4 log inactivation or reduction for viruses and bacteria
- Minimum CT factor of 12 min.mg/L and chlorine residual of 0.2 mg/L
- Disinfection by-products (trihalomethanes [THMs], haloacetic acids [HAAs], chlorite and bromate) at acceptable levels
- Acceptable colour, odour, and taste

Drinking Water Treatment for Surface Water Supplies Policy

More recently, VIHA issued the *Drinking Water Treatment for Surface Water Supplies Policy* which has stricter requirements on treatment for water systems supplied by surface water. Treatment goals for surface water systems are the following:

- 4 log inactivation or removal of viruses;
- 3 log inactivation or removal of *Cryptosporidium* and *Giardia;*
- 2 treatment processes (usually filtration and disinfection);
- 1 ntu turbidity maximum in the finished water;
- Filtration deferral may be permitted under the following conditions:
  - Turbidity be less than 1 ntu 95 percent of the time, and peak turbidity readings be less than 5 ntu for no more than 2 days in a 1-year period;
  - No more than 10 percent of raw water samples exceed 20 Escherichia coli/100 mL in any 6-month period;
  - Two primary disinfectants be used; the two together need to achieve the 4 log inactivation or reduction of viruses and 3 log inactivation or reduction of *Cryptosporidium* and *Giardia;*
  - Effective ongoing watershed protection;

As well, the VIHA may require additional treatment to address the following:

- High bacterial counts or risks of fecal contamination of source water;
- High organic matter that may result in unacceptable levels of disinfection by-products;
- Chemicals or other contaminants that may affect potability.

In 2009, The Town retained Koers and Associates to produce a Drinking Water System Assessment report which identified specific actions that the Town were to take to manage a number of potential risks associated with our water system. These risks included:

- Physical Priority Improvements;
- Water Quality Monitoring;
- Longer term Actions.

These actions have been the focus of efforts by the Town to meet the 4-3-2-1 water quality objectives laid out by VIHA since 2009. However, recent water quality test results, particularly related to turbidity at Holland Lake and recent e-coli raw water sampling may result in the Town having to amend the original recommend approach of filtration deferral through selective use of alternate supply sources, which has been the Town's practice until the present time. This will be discussed further in the next sections.

6. Water Quality Monitoring Program

The Town monitors and records water quality parameters in general compliance with the requirements of our Operational Permit issued by VIHA. In 2009, Koers and Associates issued a Water Quality Monitoring Program Report (Ref 4) that guides the collection of a number of water quality parameters for the Town, summarized as follows:

Table 4 Water Testing Frequencies

Parameter	Frequency by Location			
	Holland Lake	**Chicken Ladder Intake (Holland Creek)**	**Balancing Reservoir (Stocking Lake)**	**Distribution System**
Physical Parameters				
Turbidity	Weekly or Continuous*	Weekly or Continuous*	Weekly or Continuous*	-
Total Organic Carbon (TOC)	Monthly	Monthly	Monthly	-
Dissolved Organic carbon (DOC)	Monthly	Monthly	Monthly	-
True Color	Weekly	Weekly	Weekly	-
Apparent Color	Weekly	Weekly	Weekly	-
Temperature	Weekly	Weekly	Weekly	Along with regular health Sampling
Ultraviolet Transmissivity (UVT)	Semi-Monthly	Semi-Monthly	Semi-Monthly	-
Total Dissolved Solids (TDS)	Monthly	Monthly	Monthly	Monthly
Chemical Parameters				
Alkalinity	Monthly	Monthly	Monthly	Monthly
Physical Parameters	Monthly	Monthly	Monthly	Monthly
Calcium	Monthly	Monthly	Monthly	Monthly
Bromide	Monthly	Monthly	Monthly	-
Hardness	Semi-Annually	Semi-Annually	Semi-Annually	-
Total Metals	Semi-Annually	Semi-Annually	Semi-Annually	Semi-Annually
Microbiological Parameters				
E Coli	Semi-Annually	Semi-Annually	Semi-Annually	Semi-Annually
Total Coliforms	Semi-Annually	Semi-Annually	Semi-Annually	Semi-Annually
Regular Health Sampling	-	-	-	?
Miscellaneous Parameters				
Total Extractable Hydrocarbons (T.E.H.)	-	Following high turbidity events	-	-
THM Formation	Quarterly, or following high color events	Quarterly, or following high color events	Quarterly, or following high color events	Bi-Monthly
Chlorine Residual	-	-	-	Along with regular health Sampling
* Actual Turbidity Frequency	Continuous	Daily	Continuous	Continuous

The results of the annual lab tests (excluding turbidity and health Bacteriological testing) for Holland Lake, Stocking Lake, and Chicken Ladder Intake are detailed in Appendix F. Lab results are discussed in the following section.

6.1. Physical Parameters

a) Turbidity

The Town receives its source water from either Chicken Ladder Intake (on Holland Creek upstream of the Arbutus reservoir), or Stocking Lake. The Chicken Ladder intake includes water being released from Holland Lake, as well as additional catchment water collected downstream from Holland Dam through the Holland Creek watershed. The Town has the ability to 'switch' water supplies rapidly in response to turbidity events in either sources of supply, and hence has the ability to influence the water quality of water being delivered to the system, subject to total annual flow and time of year limitations set out by our respective water licenses. The table below provides the blended turbidity values for water that is used for the Ladysmith system. Details on the flow splits between Stocking and Chicken ladder (Holland Creek) can be found in Section 4:

Table 5 Blended Water (Either Stocking Lake, or Chicken Ladder)

Month	Ave	High	Low	Days > 1 NTU	Days > 5 NTU	Percent Stocking Water	Percent Holland Water
January	0.39	0.47	0.36	0	0	100	0
February	0.28	0.36	0.21	0	0	100	0
March	0.25	0.35	0.18	0	0	3	97
April	0.28	0.42	0.15	0	0	55	45
May	0.25	0.41	0.08	0	0	14	86
June	0.25	0.55	0.18	0	0	20	80
July	0.38	0.60	0.22	0	0	2	98
August	0.29	0.48	0.24	0	0	1	99
Sept	0.26	0.62	0.08	0	0	11	89
Oct	0.28	0.43	0.15	0	0	22	78
Nov	0.35	0.56	0.20	0	0	64	36
Dec	0.31	0.62	0.12	0	0	8	92
Total				**0**	**0**	**35**	**65**
Average	**0.30**	**0.49**	**0.18**				
Max/Min		**0.62**	**0.08**				

Our blended water easily meets the VIHA Guidelines for Surface Water Supplies for turbidity in 2013.

In addition to measuring turbidity in our 'managed' water supply, the Town also collects turbidity data for all of our sources of supply, irrespective of whether the supply is being utilized at the moment. This includes the recently installed continuously monitored turbidity meter at Holland Lake and Stocking Lakes, as well as 'manual' field turbidity samples that are taken weekly by staff at both Stocking lake, as well as our Chicken Ladder (Holland Creek) intake. The results for 2013 are as follows:

Holland Lake

The results of our first full year of continuous Holland Lake turbidity monitoring are as follows:

Table 6 Holland Lake Turbidity Summary (NTU)

Month	Ave	High	Low	Days > 1 NTU	Days > 5 NTU
January	1.320	4.532	0.383	31	0
February	1.962	3.843	0.447	25	0
March	1.742	4.223	0.700	27	0
April	0.556	2.121	0.380	0	0
May	0.699	2.294	0.353	0	0
June	0.577	0.836	0.413	0	0
July	0.740	3.146	0.401	2	0
August	0.602	1.598	0.345	0	0
Sept	0.706	1.986	0.529	1	0
Oct	0.719	1.455	0.656	0	0
Nov	0.836	2.298	0.644	4	0
Dec	0.663	1.040	0.613	0	0
Summary	**0.927**	**4.532**	**0.345**	**90**	**0**
Percent				**24.7%**	**0%**

In order for the Town to successfully argue for a filtration deferral under VIHA guidelines, Holland Lake turbidity must not exceed 1 NTU more than 5% of the year (all but 18 days), and must not exceed 5 NTU 99.5% of the year (all but 2 days). Holland Lake Turbidity did not exceed 5 NTU at any time of the year, but exceeded 1 NTU a total of 90 days in 2013, exceeding the maximum limit of 18 days by a factor of about 5 (Current data for the 2013/2014 season suggest that Holland Lake will exceed 18 days over 1 NTU this year as well).

We expect that the higher turbidity is the result of the Lake turning over during winter snow and icing conditions, but we did see some sporadic turbidity events during the summer as well.

The results shown above represent automatic turbidity recordings, however, staff also recorded manual samples on an approximately bi-weekly basis during 2013, which generally correspond to the turbidity recorder readings noted above.

The results are indicating that it is not likely that the Town could obtain a filtration deferral should the Town wish to use Holland Lake water 'directly' 100% of the time (i.e. by construction of a pipeline connecting the Town's supply pipelines to Holland Lake). It might be possible to selectively use Stocking Lake water during periods of time when the turbidity levels in Holland Lake are above 1 NTU, but this may be limited in the future by water licensing considerations at Stocking (currently under review).

Stocking Lake Turbidity

Stocking lake generally produces water under 1.0 NTU during the entire year. The Town installed a continuous turbidity meter on the Stocking lake system a number of years ago. With the exception of one isolated turbidity sample which is likely the result of a meter recoding error or external influence not related to the actual water quality, the turbidity results for 2013 are under 1.0 NTU for all of the year except for one day in June.

Table 7 Stocking Lake Continuous Turbidity Results

Month	Complete	Ave	High	Low	Days > 1 NTU	Days > 5 NTU
January	Yes	0.392	0.582	0.346	0	0
February	Yes	0.277	0.384	0.184	0	0
March	Yes	0.259	0.385	0.160	0	0
April	Yes	0.271	0.378	0.132	0	0
May	Yes	0.257	0.472	0.127	0	0
June	Yes	0.267	1.169	0.130	1	0
July	Yes	0.286	0.428	0.180	0	0
August	Yes	0.290	0.535	0.192	0	0
Sept	Yes	0.233	0.409	0.141	0	0
oct	Yes	0.267	0.669	0.088	0	0
Nov	Yes	0.412	0.587	0.152	0	0
Dec	Yes	0.346	0.745	0.213	0	0
Summary		**0.296**	**1.169**	**0.088**	**1**	**0**

Chicken Ladder Intake (Lower Holland Creek)

Staff visit Chicken Ladder intake on a daily basis, and collects a turbidity sample during that inspection. The sampling for 2013 includes a total of 376 readings and samples tested, and are summarized below:

Table 8 Holland Creek (at Chicken Ladder) Measured Turbidity Results

Month	No of Samples	Ave	High	Low	Days > 1 NTU(1)	Days > 1 NTU(2)	Days > 5 NTU
January	32	0.380	1.480	0.150	1	0	0
February	25	0.350	0.730	0.180	0	0	0
March	29	0.500	2.200	0.190	3	0	0
April	33	0.570	3.110	0.190	3	0	0
May	35	0.570	0.900	0.300	0	0	0
June	32	0.380	0.680	0.230	0	0	0
July(3)	32	0.330	1.000	0.210	1	1	0
August	34	0.290	0.490	0.240	0	0	0
Sept	30	0.420	2.970	0.210	2	0	0
Oct	33	0.280	2.610	0.180	1	0	0
Nov	29	0.460	2.400	0.200	1	0	0
Dec	32	0.340	1.390	0.150	1	0	0
Summary	**376**	**0.406**	**3.110**	**0.150**	**13**	**1**	**0**

Notes

(1): Irrespective of Source used that day

(2): When Chicken Ladder used as the source water

(3): Anomaly in a test result - no rain month of July

On an average daily sample basis (i.e. only one sample per day), it appears that in 2013, Chicken Ladder intake had lower turbidity levels than Holland Lake. This is probably due to the lack of major rain and/or slide events in the lower Holland watershed in 2013 – we would normally expect to see more variation in Chicken Ladder intake turbidity due to the relatively small reservoir capacity at that location (only a few minutes of storage is available at Chicken Ladder intake).

Staff monitor turbidity levels at Chicken Ladder on a regular basis, and are able to 'switch' water supplies rapidly when turbidity events occur (usually the result of high intensity rain storms occurring in the upper Holland watershed). The table above shows that in 12 out of 13 instances where turbidity rose above 1.0, staff had already switched the water supply over to Stocking Lake. In the one instance (July 4th, 2013), we noted a field reading of 0.4 NTU, and the lab result came back a week later at 1.0 NTU, so we suspect that this was an erroneous result (it did not rain the entire month of July, 2013).

If the Holland Lake turnover situation (Jan-March, 2013) is ignored, Holland Lake continues to produce slightly lower turbidity water than Chicken Ladder (7 days over 1 NTU vs 13 days at Chicken Ladder). However, we know from previous year's data collection that Chicken ladder is subject to rapid turbidity increases due to sudden rainstorms in the watershed, and as such, the relatively few days over 1 NTU for 2013 should not be considered a normal occurrence.

b) Organic Carbon, Color, Temperature, Ultraviolet Transmissivity, Dissolved Solids

Results for Holland Lake, Stocking Lake, and Chicken Ladder are attached in Appendix B, and summarized as flows:

Table 9 Holland Lake – Other Physical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Total Dissolved Solids	10	Mg/L	19.6	27.0	13.0	<500	OK
PH	12	PH Units	6.38	6.60	6.20	6.5-8.4	OK
True Color	44	Col Unit	10.36	15.0	5.0	<15 (AO)	OK
Apparent Color	43	Col Unit	30.05	120.0	10.0		
Dissolved organic Carbon	9	Mg/L	2.55	2.98	2.09	<5 (AO)	OK
Total Organic Carbon	9	Mg/L	3.02	4.01	2.22	n/a	
Ultraviolet Transmissivity	20	AU/cm	76.61	85.2	51.0	>80 (AO)	OK – review for UV Disinfection

(AO) Aesthetic Objective

Table 10 Holland Creek (Chicken Ladder) – Other Physical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Total Dissolved Solids	14	Mg/L	30.5	62.0	18.0	<500	OK
PH	12	PH Units	6.77	6.90	6.60	6.5-8.4	OK
True Color	49	Col Unit	12.76	30.0	5.0	<15 (AO)	OK
Apparent Color	49	Col Unit	15.76	40.0	10.0		
Dissolved organic Carbon	11	Mg/L	2.46	5.34	1.14	<5 (AO)	OK (marginally)
Total Organic Carbon	11	Mg/L	2.71	5.65	1.78	n/a	
Ultraviolet Transmissivity	24	AU/cm	80.02	88.1	64.3	>80 (AO)	OK – review for UV Disinfection

(AO) Aesthetic Objective

Table 11 Stocking Lake – Other Physical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Total Dissolved Solids	12	Mg/L	29.75	50.0	13.0	<500	OK
PH	12	PH Units	6.98	7.10	6.70	6.5-8.4	OK
True Color	51	Col Unit	9.6	14.0	5.0	<15 (AO)	OK
Apparent Color	51	Col Unit	12.82	19.0	5.0		
Dissolved organic Carbon	11	Mg/L	2.30	3.13	1.65	<5 (AO)	OK
Total Organic Carbon	11	Mg/L	2.63	3.92	1.93	n/a	
Ultraviolet Transmissivity	25	AU/cm	84.8	88.7	78.5	>80 (AO)	OK

(AO) Aesthetic Objective

The results meet the Canadian Drinking Water Guidelines, although some results, listed as aesthetic guidelines, may have a bearing on the design of water treatment facilitates should they be considered in the future.

6.2. Chemical Parameters

The Town is required to conduct bi-annual metals testing. Results for all three water sources are as follows in Table 12. All three water sources meet the Canadian Drinking Water Guidelines.

Table 12 Metals, Hardness Testing – June 20, 2013

Maxxam ID		GS1922	GS1923	GS1924		
Sampling Date		2013-06-20 10:42	2013-06-20 8:00	2013-06-20 8:05		
COC Number		08367831	08367831	08367831		
	UNITS	**HOLLAND LAKE**	**STOCKING LAKE**	**CHICKEN LADDER**	**CDW/BC Guideline**	**Comments**
Calculated Parameters						
Total Hardness (CaCO3)	mg/L	3.61	10.7	7.28	0.50	OK
Total Metals by ICPMS						
Total Aluminum (Al)	ug/L	58.5	37.6	53.0	100	OK
Total Antimony (Sb)	ug/L	<0.50	<0.50	<0.50	6	OK
Total Arsenic (As)	ug/L	<0.10	0.13	0.13	5	OK
Total Barium (Ba)	ug/L	3.3	3.3	3.6	1000	OK
Total Beryllium (Be)	ug/L	<0.10	<0.10	<0.10		
Total Bismuth (Bi)	ug/L	<1.0	<1.0	<1.0		
Total Boron (B)	ug/L	<50	<50	<50	500	OK
Total Cadmium (Cd)	ug/L	0.041	0.094	0.039	5	OK

Table 12 (Cont'd)
Metals, Hardness Testing – June 20, 2013

Maxxam ID		GS1922	GS1923	GS1924		
Sampling Date		2013-06-20 10:42	2013-06-20 8:00	2013-06-20 8:05		
COC Number		08367831	08367831	08367831		
	UNITS	**HOLLAND LAKE**	**STOCKING LAKE**	**CHICKEN LADDER**	**C,BC DW Guideline**	**Comments**
Total Chromium (Cr)	ug/L	<1.0	<1.0	<1.0	50	OK
Total Cobalt (Co)	ug/L	<0.50	<0.50	<0.50		
Total Copper (Cu)	ug/L	1.70	**2.28**	1.39	2	Stocking Over
Total Iron (Fe)	ug/L	189	70.9	54.9	300 AO	OK
Total Lead (Pb)	ug/L	<0.20	0.22	<0.20	1	OK
Total Lithium (Li)	ug/L	<5.0	<5.0	<5.0		
Total Manganese (Mn)	ug/L	19.2	3.0	1.6	50 AO	OK
Total Mercury (Hg)	ug/L	<0.050	<0.050	<0.050	1	OK
Total Molybdenum (Mo)	ug/L	<1.0	<1.0	<1.0	73	
Total Nickel (Ni)	ug/L	<1.0	<1.0	<1.0		
Total Selenium (Se)	ug/L	<0.10	<0.10	<0.10	2	OK
Total Silicon (Si)	ug/L	640	1510	2450		
Total Silver (Ag)	ug/L	<0.020	<0.020	<0.020		
Total Strontium (Sr)	ug/L	5.9	11.0	12.2		
Total Thallium (Tl)	ug/L	<0.050	<0.050	<0.050		
Total Tin (Sn)	ug/L	<5.0	<5.0	<5.0		
Total Titanium (Ti)	ug/L	<5.0	<5.0	<5.0		
Total Uranium (U)	ug/L	<0.10	<0.10	<0.10		
Total Vanadium (V)	ug/L	<5.0	<5.0	<5.0		
Total Zinc (Zn)	ug/L	<5.0	<5.0	<5.0	7.5	OK
Total Zirconium (Zr)	ug/L	<0.50	<0.50	<0.50		
Total Calcium (Ca)	mg/L	1.08	3.45	2.22		
Total Magnesium (Mg)	mg/L	0.218	0.500	0.422		
Total Potassium (K)	mg/L	0.124	0.276	0.100		
Total Sodium (Na)	mg/L	0.656	1.16	1.18	< 200 AO	OK
Total Sulphur (S)	mg/L	<3.0	<3.0	<3.0		

6.3. Other Chemical Parameters

The Town also tests for Alkalinity, Calcium, Bromide, and Hardness. Results of these tests are as follows:

Table 13 Holland Lake - Other Chemical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Alkalinity	12	Mg/L	3.59	6.50	2.40	n/a	Slightly low
Calcium	12	Mg/L	1.31	1.78	0.95	n/a	
Bromide	12	Mg/L	0.00	0.00	0.00	n/a	

Table 14 Holland Creek (Chicken Ladder) - Other Chemical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Alkalinity	12	Mg/L	4.91	6.60	3.30	n/a	Slightly low
Calcium	12	Mg/L	1.94	2.73	1.19	n/a	
Bromide	12	Mg/L	0.00	0.00	0.00	n/a	

Table 15 Stocking Lake - Other Chemical Results

Parameter	No of Samples	Units	Ave	High	Low	CDW Guideline	Comments
Alkalinity	12	Mg/L	8.83	10.60	7.90	n/a	Slightly low
Calcium	12	Mg/L	3.52	4.73	1.33	n/a	
Bromide	12	Mg/L	0.00	0.00	0.00	n/a	

6.4.Microbiological Parameters

The Town, through Island Health, conducts weekly tests for E.Coli and Total Coliforms of our distribution system (treated water). Tests are taken weekly, and are enclosed in Appendix E. The Town is required to meet the following standards set out by Island Health for our distribution system:

Parameter	Standard
Fecal Coliform Bacteria	No detectable fecal coliform bacteria per 100 ml
Escherichia Coli	No detectable Escherichia Coli per 100 ml
Total Coliform Bacteria	At least 90% of samples have no detectable total coliform bacteria per 100 ml and no sample has more than 10 total coliform

The Town also tests our raw water sources for E-Coli and Total Coliforms. The E-Coli tests are used to determine generally eligibility for filtration deferral – in any 6 month period, no more than 10% of raw water samples can test over 20 cfu/100ml. Results for 2013 are summarized as follows:

Table 16 Source Water E-Coli and Total Coliform Results Holland Lake

Parameter	No of Samples	Units	Ave	High	Low	Samples > 20 (e-coli)	Comments
E-Coli	19	Cfu/100ml	3.5	15	1	0.0% > 20 cfu/100ml	10% max
Total Coliforms	19	Cfu/100ml	325	870	36	68% > 100 cfu/100ml	

Table 17 Source Water E-Coli and Total Coliform Results Holland Creek (Chicken Ladder)

Parameter	No of Samples	Units	Ave	High	Low	Samples > 20 (e-coli)	Comments
E-Coli	23	Cfu/100ml	5.63	21	1	4.3% > 20 cfu/100ml	10% max
Total Coliforms	23	Cfu/100ml	652	1700	36	65% >100 cfu/100ml	

Table 18 Source Water E-Coli and Total Coliform Results Stocking Lake

Parameter	No of Samples	Units	Ave	High	Low	Samples > 20 (e-coli)	Comments
E-Coli	23	Cfu/100ml	1.5	3	1	0% > 20 cfu/100ml	10% max
Total Coliforms	23	Cfu/100ml	347	2400	5	34% > 100 cfu/100ml	

For 2013, approximately 5% of the raw E-Coli tests for Chicken Ladder exceeded the maximum allowable E-Coli value of 20 cfu/100ml. This marginally meets the requirement for filtration deferral under VIHA's filtration deferral policy, although Chicken Ladder had 14% of E-Coli tests over 20 cfu/100ml in 2012.

6.5. Miscellaneous Parameters

THM Formation

Trihalomethanes (THM's) and Haloacetic Acids (HAA's) are formed as products of conventional chlorination within a water system. They are commonly referred to disinfection by-products. The Canadian Drinking Water guideline is 0.1 mg/L for THM's and 0.08 mg/L for HAA's.

A total of 8 tests were conducted within the water system in 2013 for each, as follows:

Table 19 THM Formation in Water System, in mg/L

	558 Hooper	1280 Rocky Creek	CDW Guideline	Comments
Date	**THM**	**THM**		
Mar.7/13	0.078	0.070	0.100	Marginally meets CDWG
Jun.17/13	0.068	0.069	0.100	Marginally meets CDWG
Sept.18/13	0.075	0.065	0.100	Marginally meets CDWG
Dec.11/13	0.070	0.058	0.100	Marginally meets CDWG

Table 20 Total HAA Formation in Water System, in mg/L

	558 Hooper	1280 Rocky Creek	CDW Guideline	Comments
Mar 7, 13	0.091	0.085	0.080	Marginally Fails CDWG
Jun.17/13	0.120	0.110	0.080	Marginally Fails CDWG
Sept.18/13	0.102	0.086	0.080	Marginally Fails CDWG
Dec.11/13	0.130	0.090	0.080	Marginally Fails CDWG

Table 21 HAA Component Results, in mg/L

Test Date	Component	558 Hooper	1280 Rocky Creek
Mar 7, 13	Chloriform	0.076	0.068
Mar 7, 13	Chlorodibromomethane	<0.001	<0.001
Mar 7, 13	Bromodichloromethane	.0022	0.0020
Mar 7, 13	Bromoform	<0.001	<0.001
Mar 7, 13	1,4-Difluorobenzene (sur.)	0.099	0.101
Mar 7, 13	4-Bromofluorobenzene (sur.)	0.088	0.095
Mar 7, 13	D4-1,2-Dichloroethane (sur.)	0.110	0.113
June 17, 2013	Chloriform	0.066	0.067
June 17, 2013	Chlorodibromomethane	<0.001	<0.001
June 17, 2013	Bromodichloromethane	.0019	0.0018
June 17, 2013	Bromoform	<0.001	<0.001
June 17, 2013	1,4-Difluorobenzene (sur.)	0.099	0.104
June 17, 2013	4-Bromofluorobenzene (sur.)	0.081	0.094
June 17, 2013	D4-1,2-Dichloroethane (sur.)	0.082	0.095
Sept 18, 2013	Chloriform	0.071	0.062
Sept 18, 2013	Chlorodibromomethane	<0.001	<0.001
Sept 18, 2013	Bromodichloromethane	.0036	0.0032
Sept 18, 2013	Bromoform	<0.001	<0.001
Sept 18, 2013	1,4-Difluorobenzene (sur.)	0.103	0.104
Sept 18, 2013	4-Bromofluorobenzene (sur.)	0.089	0.092
Sept 18, 2013	D4-1,2-Dichloroethane (sur.)	0.104	0.105

It should be noted that the HAA testing for 2012 also marginally did not meet current Canadian Drinking Water Quality Guidelines.

Health Canada has published an extensive review of the effects of Haloacetic Acids in drinking water, it can be accessed at:

http://www.hc-sc.gc.ca/ewh-semt/pubs/water-eau/haloaceti/index-eng.php

A copy of the executive summary follows, and provides an overview of this issue:

"Haloacetic acids (HAAs) are a group of compounds that can form when the chlorine used to disinfect drinking water reacts with naturally occurring organic matter (e.g., decaying leaves and vegetation). The use of chlorine in the treatment of drinking water has virtually eliminated waterborne diseases, because chlorine can kill or inactivate most microorganisms commonly found in water. The majority of drinking water treatment plants in Canada use some form of chlorine to disinfect drinking water: to treat the water directly in the treatment plant and/or to maintain a chlorine residual in the distribution system to prevent bacterial regrowth. Disinfection is an essential component of public drinking water treatment; the health risks from disinfection by-products, including haloacetic acids, are much less than the risks from consuming water that has not been appropriately disinfected.

The haloacetic acids most commonly found in drinking water are monochloroacetic acid (MCA), dichloroacetic acid (DCA), trichloroacetic acid (TCA), monobromoacetic acid (MBA) and dibromoacetic acid (DBA). Of these, DCA and TCA have been most extensively studied, and there are some scientific data available on MCA and DBA. However, insufficient data were available to allow the development of an individual guideline for MBA.

This Guideline Technical Document reviews the health risks associated with haloacetic acids in drinking water. It assesses all identified health risks, taking into account new studies and approaches, as well as treatment considerations. Exposure to haloacetic acids from drinking water through inhalation and skin contact has been considered for inclusion, but is not deemed significant. Based on this review, the guideline for total haloacetic acids in drinking water is established at a maximum acceptable concentration of 0.08 mg/L. This guideline takes into consideration the availability of appropriate treatment technologies and the ability of treatment plants to meet the guideline without compromising the effectiveness of disinfection."

With respect to the impact of water treatment on HAA values, the study notes that:

"The removal of organic precursors is the most effective way to reduce the concentrations of all DBPs, including HAAs, in finished water (U.S. EPA, 1999c; Reid Crowther & Partners Ltd., 2000). These precursors include synthetic organic compounds and NOM, which can react with disinfectants to form HAAs. Removing HAA precursors will also result in the formation of lower concentrations of HAAs (Reid Crowther & Partners Ltd., 2000). Conventional municipal-scale water treatment techniques (coagulation, sedimentation, dissolved air flotation, precipitative softening and filtration) can reduce the amount of HAA precursors, but are ineffective in removing HAAs once they are formed."

We expect that the removal of organic compounds, likely found in the greatest concentrations at Holland Creek (Chicken ladder), will effectively by dealt with through additional treatment, including filtration.

7. Watershed Management

Both Holland and Stocking Lake watersheds are jointly used 'semi-closed' watersheds, owned and managed by a number of parties, including:

- Town of Ladysmith;
- Ministry of Forests, Lands and Natural Resource Operations (MFNRO);
- TimberWest Forest Corporation (TimberWest);
- Cowichan Valley Regional District (CVRD).

MFNRO leases some of the lands to private third parties and First Nations principally for forestry use.

There are not any public roads within the watershed area, but are predominately private forest access roads that are jointly used by the parties.

The public is encouraged to use a recreational trail system within the watershed, and includes trails that are adjacent to both Holland and Stocking Lakes. The Town maintains a well-used trail system that includes the above noted trail routes. Signs are posted at a number of strategic locations prohibiting vehicle access to the lakes, and recreational lake use such as boating, swimming, and fishing is prohibited.

7.1. Security

The Town jointly manages a number of gates within the Holland and Stocking lakes watersheds that control access into Holland and Stocking Lakes. These gates are normally left in a locked state, except when active logging activities are taking place. Town staff travel to both Lakes on a minimum weekly basis.

In recent years, the Town has observed instances where 4 wheel ATV vehicles have entered the area by by-passing the locked gates, and the Town and Timberwest has been working to prevent this activity through the use of ditching and other means to block access into the area, with some success.

The Town also posts signs at entrances to both lakes advising that public access to the lakes is not permitted. The Town continues to see evidence of occasional camping and recreational activity around the lakes, particularly Holland lake which is more remote than Stocking lake.

The Town has been working with the various stakeholders of the southern access area (South Watts Road) to improve the gate at that location, possibly through the use of access cards and video security, as this gate is located nearer a source of power. Funds have been allocated in the capital budget for this purpose.

8. Water System Inquiries and Complaints

In 2012, staff at Public Works created a task tracking system for incoming requests for service. The following table summarizes requests for service that were received in 2013, sorted by general complaint type:

Table 22 – 2013 Requests for Service

Task ID	Date received	Issue	Meter	Leak	Quality	Pressure	Other	Status
116	07-Mar-13	Color			x			Complete
129	20-Mar-13	Leak		x				Complete
190	08-May-13	Meter	x					Complete
221	05-May-13	Meter	x					Complete
258	17-May-13	Pressure				x		Complete
267	28-May-13	Break		x				Complete
408	29-Jul-13	Leak at meter		x				Complete
422	31-Jul-13	Color			x			Complete
442	13-Aug-13	Leak		x				Complete
458	19-Aug-13	Leak		x				Complete
465	20-Aug-13	Pressure				x		Complete
494	03-Sep-13	Non Metered Usage					x	Complete
496	03-Sep-13	Leak		x				Complete
509	09-Sep-13	Meter new install	x					Complete
511	10-Sep-13	Leak		x				Complete
512	11-Sep-13	Meter	x					Complete
529	18-Sep-13	Leak (private)		x				Complete
567	07-Oct-13	Leak		x				Complete
Totals			**4**	**9**	**2**	**2**	**1**	

All requests for service were completed within 2 weeks.

9. Routine Maintenance Program

The Town has a regular maintenance program that is described briefly as follows:

9.1.Distribution

- Water mains are flushed using a unidirectional flushing program
- Air relief valves are cleaned
- Fireline meters are cleaned
- Fire Hydrants are completely disassembled and inspected on a 2 year rotation
- Paint and brush out around hydrants as needed
- All irrigation backflow prevention devices tested and repaired if needed

9.2.Source Intakes

- Winter maintenance of chlorination system while off line
- Weekly blowing of air lines through intake screens
- Daily checks of pump flows and chlorine levels
- Monthly calibration of turbidity analyzers

9.3.Reservoirs

- Daily security check of tanks and compounds
- Yearly cleaning
- Clean Reservoir using divers every 5 years.

9.4.Pump Stations

- Daily checks of pumps and chlorination system
- Security checks of compounds
- Bi-Annual calibration of chlorine analyzers and turbidimeters

The Town trains it's staff in accordance with current industry practises and requirements, and staff are certified to work on our system as required in the Town's Operating Permit.

10. Capital Planning

The Town is in the process of updating our water capital plan, particularly as it relates to water filtration. A copy of the 'current' (at end of 2013) detailed long term capital plan is enclosed in Appendix C. A summary of the major capital projects projected over the next 5 years is as follows:

Table 23 – Long Term Capital Plan

Title	Description	Purpose	Start Year	Cost
Chlorination Facility	Construction of a new Chlorination Facility	To upgrade the present chlorination facility.	2013	$1,500,000
Stocking Lake Supply Main Replacement - Phase I	Replacement of the existing AC watermain from Stocking lake to the balancing reservoir.	To provide a more secure Supply main.	2015	$800,000
Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.	2015	$300,000
Holland to Stocking Supply Main	New Supply main to connect Stocking Supply Main with Holland Lake.	To provide a direct system connection from Holland Lake to the Town's water system.	2016	$4,000,000
Stocking Lake Supply Main Replacement - Phase II	Replacement of the AC Supply watermain from balancing reservoir to the Old Chlorination Station.	To provide a more secure Supply main.	2016	$800,000
Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	2016	$6,265,000
Chicken Ladder Intake Upgrade	Replacement and Upgrade of Chicken Ladder Intake (Lower Holland Creek).	To improve reliability and safety of water supply.	2017	$2,000,000
Chicken Ladder Supply Main replacement	Replacement of 1960's AC supply main from Chicken Ladder to Water Treatment Plant.	To provide a more secure Supply main.	2017	$1,000,000
Energy Recovery Plant - Old PRV Station	Construction of an Energy recovery Plant	Recover Energy Costs from pressure reduction in supply main	2017	$650,000
Replace Holland Supply Main: Pub Works Yard to Colonia	Replacement of old AC supply main along the Holland Creek Trail	To provide a more secure Supply main.	2017	$330,000
Arbutus reservoir - Capacity Increase	Doubling of Arbutus Reservoir	For future development	2018	$720,000
Energy Recovery Plant - Stocking Lake	Construction of an Energy recovery Plant	Recover Energy Costs from pressure reduction in supply main	2018	$570,000
New Upper Pressure Zone Reservoir	To service Couverdon, possibly others.	For development in upper zone	2018	$520,000
Replace Supply Main: Colonia to Treatment Plant		To provide a more secure Supply main.	2018	$200,000
Upper Pressure Zone - Pump Station		For development in upper zone	2018	$670,000
			5 year Plan - Cost of "Major' Projects Only	**$20,325,000**

Appendix A – Canadian Drinking Water Quality Guidelines

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 1 – Microbiological Parameters

Parameter (approval)	Guideline	Common sources	Health considerations	Applying the guideline
Bacterial waterborne pathogens (2006)	None required	Human and animal faeces; some are naturally occurring	Commonly associated with gastrointestinal upset (nausea, vomiting, diarrhoea); some pathogens may infect the lungs, skin, eyes, central nervous system or liver.	Use multi-barrier approach to reduce pathogens to levels that are non-detectable or not associated with illness.
Enteric viruses (2011)	Treatment goal: Minimum 4 log reduction and/or inactivation of enteric viruses	Human and animal faeces	Commonly associated with gastrointestinal upset (nausea, vomiting, diarrhoea); less common health effects can include respiratory symptoms, central nervous system infections, liver infections and muscular syndromes.	Routine monitoring for viruses is not practical; where possible, characterize source water to determine if greater than a 4 log removal or inactivation is necessary.
Escherichia coli(*E. coli*) (2006)	MAC: None detectable per 100 mL	Human and animal faeces	The presence of *E. coli* indicates recent faecal contamination and the potential presence of microorganisms capable of causing gastrointestinal illnesses; pathogens in human and animal faeces pose the most immediate danger to public health.	*E. coli* is used as an indicator of the microbiological safety of drinking water; if detected, enteric pathogens may also be present.
Heterotrophic plate count (HPC) (2006)	None required	Naturally occurring	HPC results are not an indicator of water safety and should not be used as an indicator of potential adverse human health effects; HPC is a useful operational tool for monitoring general bacteriological water quality through the treatment process and in the distribution system.	If increases in HPC values above baseline levels occur, the system should be inspected to determine the cause; HPC should be minimized through effective treatment and disinfection and remain constant over time.
Protozoa:*Giardia* and*Cryptosporidium*(2004)	Treatment goal: Minimum 3 log reduction and/or inactivation	Human and animal faeces	Commonly associated with gastrointestinal upset (nausea, vomiting, diarrhoea); less common health effects can include respiratory symptoms, central nervous system infections, liver infections and muscular syndromes.	Monitoring for *Cryptosporidium* and*Giardia* in source waters will provide valuable information for assessing treatment requirements.

Guidelines for Canadian Drinking Water Quality (Aug 2012) Table 1 – Microbiological Parameters				
Parameter (approval)	**Guideline**	**Common sources**	**Health considerations**	**Applying the guideline**
Total coliforms (2006)	*At exit of municipal treatment plant or throughout semi-public systems*: MAC of none detectable/100 mL *In municipal distribution systems*: No consecutive samples or no more than 10% of samples should contain total coliforms	Human and animal faeces; naturally occurring in water, soil and vegetation	Total coliforms are not used as indicators of potential health effects from pathogenic microorganisms; they are used as an operational tool to determine how well the drinking water treatment system is operating.	In water leaving a treatment plant, the presence of total coliforms indicates that the water has been inadequately treated and may contain pathogenic microorganisms; in semi-public systems, the presence of total coliforms generally indicates that the system is vulnerable to contamination and that additional actions need to be taken; in a distribution and storage system, detection of total coliforms can indicate regrowth of the bacteria in distribution system biofilms or intrusion of untreated water; thus, exceedances of the distribution system goal should be investigated.
Turbidity (2003)	Guideline Treated water < 0.1 NTUTable 1 footnote1at all times. Where not achievable: ≤ 0.3 NTUTable 1 footnote2 ≤ 1.0 NTUTable 1 footnote3 ≤ 0.1 NTUTable 1 footnote4	Naturally occurring particles: *Inorganic*: clays, silts, metal precipitates *Organic*: decomposed plant & animal debris, microorganisms	Indirect associations: particles can harbour microorganisms, protecting them from disinfection, and can entrap heavy metals and biocides; elevated or fluctuating turbidity in filtered water can indicate a problem with the water treatment process and a potential increased risk of pathogens in treated water.	Guidelines apply to individual filter turbidity for systems that use surface water or GUDI; drinking water from some sources may meet exemption criteria from filtration requirements established by the appropriate authority; increases in distribution system turbidity can be indicative of deteriorating water quality and should be investigated.

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
T	Aluminum (1998)		OG: < 0.1 (conventional treatment); < 0.2 (other treatment types)	Aluminum salts used as coagulants in drinking water treatment; naturally occurring		Current weight of evidence does not indicate adverse health effects at levels found in drinking water.
I	Ammonia (1987)	None required		Naturally occurring; released from agricultural or industrial wastes; added as part of chloramination for drinking water disinfection		Guideline value not necessary as it is produced in the body and efficiently metabolized in healthy people; no adverse effects at levels found in drinking water.
I	Antimony (1997)	0.006		Naturally occurring (erosion); soil runoff; industrial effluents; leaching from plumbing materials and solder	**Health basis of MAC:**Microscopic changes in organs and tissues (thymus, kidney, liver, spleen, thyroid)	MAC takes into consideration analytical achievability; plumbing should be thoroughly flushed before water is used for consumption.
I	Arsenic (2006)	0.01 ALARA		Naturally occurring (erosion and weathering of soils, minerals, ores)	**Health basis of MAC:**Cancer (lung, bladder, liver, skin) (classified as human carcinogen) **Other:** Skin, vascular and neurological effects (numbness and tingling of extremities)	MAC based on treatment achievability; elevated levels associated with certain groundwaters; levels should be kept as low as reasonably achievable.
I	Asbestos (1989, 2005)	None required		Naturally occurring (erosion of asbestos minerals and ores); decay of asbestos-cement pipes		Guideline value not necessary; no evidence of adverse health effects from exposure through drinking water.
P	Atrazine (1993)	0.005		Leaching and/or runoff from agricultural use	**Health basis of MAC:**Developmental effects (reduced body weight of offspring) **Other:** Potential increased risk of ovarian cancer or lymphomas (classified as possible carcinogen)	MAC applicable to the sum of atrazine and its*N*-dealkylated metabolites; persistent in source waters.

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
P	Azinphos-methyl (1989, 2005)	0.02		Leaching and/or runoff from agricultural use	**Health basis of MAC:**Neurological effects (plasma cholinesterase)	All uses to be phased out by 2012.
I	Barium (1990)	1		Naturally occurring; releases or spills from industrial uses	**Health basis of MAC:**Increases in blood pressure, cardiovascular disease	
O	Benzene (2009)	0.005		Releases or spills from industrial uses	**Health basis of MAC:** Bone marrow (red and white blood cell) changes and cancer (classified as human carcinogen) **Other:** Blood system and immunological responses	MAC considers additional exposure through showering and bathing; drinking water is generally a minor source of exposure.
O	Benzo[*a*]pyrene (1988, 2005)	0.000 01		Leaching from liners in water distribution systems	**Health basis of MAC:**Stomach tumours (classified as probable carcinogen)	
I	Boron (1990)	5		Naturally occurring; leaching or runoff from industrial use	**Health basis of MAC:**Reproductive effects (testicular atrophy, spermatogenesis) **Other:** Limited evidence of reduced sexual function in men	MAC based on treatment achievability.
DBP	Bromate (1998)	0.01		By-product of drinking water disinfection with ozone; possible contaminant in hypochlorite solution	**Health basis of MAC:** Renal cell tumours (classified as probable carcinogen)	MAC based on analytical and treatment achievability
P	Bromoxynil (1989, 2005)	0.005		Leaching or runoff from agricultural use	**Health basis of MAC:**Reduced liver to body weight ratios	
I	Cadmium (1986, 2005)	0.005		Leaching from galvanized pipes, solders or black polyethylene pipes; industrial and municipal waste	**Health basis of MAC:**Kidney damage and softening of bone	
I	Calcium (1987, 2005)	None required		Naturally occurring (erosion and weathering of soils, minerals, ores)		Guideline value not necessary, as there is no evidence of adverse health effects from calcium in drinking water; calcium contributes to hardness

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
P	Carbaryl (1991, 2005)	0.09		Leaching or runoff from agricultural use	**Health basis of MAC:**Decreased kidney function (may be rapidly reversible after exposure ceases)	
P	Carbofuran (1991, 2005)	0.09		Leaching or runoff from agricultural use	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition) and growth suppression	
O	Carbon tetrachloride (2010)	0.002		Industrial effluents and leaching from hazardous waste sites	**Health basis of MAC:** Liver toxicity **Other:** Kidney damage; liver tumours (classified as probable carcinogen)	MAC considers additional exposure through showering and bathing
D	Chloramines (1995)	3		Monochloramine is used as a secondary disinfectant; formed in presence of both chlorine and ammonia	**Health basis of MAC:**Reduced body weight gain **Other:** immunotoxicity effects	MAC is for total chloramines based on health effects associated with monochloramine and analytical achievability
DBP	Chlorate (2008)	1		By-product of drinking water disinfection with chlorine dioxide; possible contaminant in hypochlorite solution	**Health basis of MAC:**Thyroid gland effects (colloid depletion)	Formation of chlorate ion should be prevented, as it is difficult to remove once formed; chlorate formation should be controlled by respecting the maximum feed dose of 1.2 mg/L of chlorine dioxide and managing /monitoring formation in hypochlorite solutions.
I	Chloride (1979, 2005)		AO: ≤ 250	Naturally occurring (seawater intrusion); dissolved salt deposits, highway salt, industrial effluents, oil well operations, sewage, irrigation drainage, refuse leachates		Based on taste and potential for corrosion in the distribution system

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
D	Chlorine (2009)	None required		Used as drinking water disinfectant	Guideline value not necessary due to low toxicity at concentrations found in drinking water	Free chlorine concentrations in most Canadian drinking water distribution systems range from 0.04 to 2.0 mg/L
D	Chlorine dioxide (2008)	None required		Used as drinking water disinfectant	A guideline for chlorine dioxide is not required because of its rapid reduction to chlorite in drinking water	A maximum feed dose of 1.2 mg/L of chlorine dioxide should not be exceeded to control the formation of chlorite and chlorate
DBP	Chlorite (2008)	1		By-product of drinking water disinfection with chlorine dioxide	**Health basis of MAC:**Neurobehavioural effects (lowered auditory startle amplitude, decreased exploratory activity), decreased absolute brain weight, altered liver weights	Chlorite formation should be controlled by respecting the maximum feed dose of 1.2 mg/L of chlorine dioxide and managing /monitoring formation in hypochlorite solutions.
P	Chlorpyrifos (1986)	0.09		Leaching and/or runoff from agricultural or other uses	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	Not expected to leach significantly into groundwater
I	Chromium (1986)	0.05		Naturally occurring (erosion of minerals); releases or spills from industrial uses	**Health basis of MAC:**Enlarged liver, irritation of the skin, respiratory and gastrointestinal tracts from chromium (VI)	Chromium (III) is an essential element; MAC is protective of health effects from chromium (VI)
T	Colour (1979, 2005)		AO: ≤ 15 TCU	Naturally occurring organic substances, metals; industrial wastes		May interfere with disinfection; removal is important to ensure effective treatment
I	Copper (1992)		AO: ≤ 1.0	Naturally occurring; leaching from copper piping	Copper is an essential element in human metabolism. Adverse health effects occur at levels much higher than the aesthetic objective	Based on taste, staining of laundry and plumbing fixtures; plumbing should be thoroughly flushed before water is used for consumption
I	Cyanide (1991)	0.2		Industrial and mining effluents; release from organic compounds	**Health basis of MAC:** No clinical or other changes at the highest dose tested	Health effects from cyanide are acute; at low levels of exposure, it can be detoxified to a certain extent in the human body

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
O	Cyanobacterial toxins--Microcystin-LR (2002)	0.0015		Naturally occurring (released from blooms of blue-green algae)	**Health basis of MAC:** Liver effects (enzyme inhibitor) **Other:** Classified as possible carcinogen	MAC is protective of total microcystins; avoid algicides like copper sulphate, as they may cause toxin release into water
P	Diazinon (1986, 2005)	0.02		Runoff from agricultural or other uses	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	Not expected to leach significantly into groundwater
P	Dicamba (1987, 2005)	0.12		Leaching or runoff from agricultural or other uses	**Health basis of MAC:** Liver effects (vacuolization, necrosis, fatty deposits and liver weight changes)	Readily leaches into groundwater
O	1,2-DichlorobenzeneTable 2 footnote2(1987)	0.2	AO: ≤ 0.003	Releases or spills from industrial effluents	**Health basis of MAC:**Increased blood cholesterol, protein and glucose levels	AO based on odour; levels above the AO would render drinking water unpalatable
O	1,4-DichlorobenzeneTable 2 footnote2(1987)	0.005	AO: ≤ 0.001	Releases or spills from industrial effluents; use of urinal deodorants	**Health basis of MAC:**Benign liver tumours and adrenal gland tumours (classified as probable carcinogen)	AO based on odour; levels above the AO would render drinking water unpalatable
O	1,2-Dichloroethane (1987)	0.005		Releases or spills from industrial effluents; waste disposal	**Health basis of MAC:**Cancer of the circulatory system (classified as probable carcinogen)	MAC based on treatment and analytical achievability
O	1,1-Dichloroethylene (1994)	0.014		Releases or spills from industrial effluents	**Health basis of MAC:** Liver effects (fatty changes)	
O	Dichloromethane (2011)	0.05		Industrial and municipal wastewater discharges	**Health basis of MAC:** Liver effects (liver foci and areas of cellular alteration). **Other:** Classified as probable carcinogen	MAC is protective of carcinogenic effects and considers additional exposure through showering and bathing
O	2,4-Dichlorophenol (1987, 2005)	0.9	AO: ≤ 0.0003	By-product of drinking water disinfection with chlorine; releases from industrial	**Health basis of MAC:** Liver effects (cellular changes)	AO based on odour; levels above the AO would render drinking water unpalatable

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
				effluents		
P	2,4-Dichlorophenoxy acetic acid (2,4-D) (1991)	0.1		Leaching and/or runoff from use as a weed controller; releases from industrial effluents	**Health basis of MAC:**Kidney effects (tubular cell pigmentation)	
P	Diclofop-methyl (1987, 2005)	0.009		Leaching and/or runoff from use as a weed controller; added directly to water to control aquatic weeds	**Health basis of MAC:** Liver effects (enlargement and enzyme changes)	Low potential for groundwater contamination
P	Dimethoate (1986, 2005)	0.02		Leaching and/or runoff from residential, agricultural and forestry use	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	
P	Diquat (1986, 2005)	0.07		Leaching and/or runoff from agricultural use; added directly to water to control aquatic weeds	**Health basis of MAC:**Cataract formation	Unlikely to leach into groundwater
P	Diuron (1987, 2005)	0.15		Leaching and/or runoff from use in controlling vegetation	**Health basis of MAC:**Weight loss, increased liver weight and blood effects	High potential to leach into groundwater
O	Ethylbenzene (1986, 2005)		AO: ≤ 0.0024	Emissions, effluents or spills from petroleum and chemical industries		Based on odour
I	Fluoride (2010)	1.5		Naturally occurring (rock and soil erosion); may be added to promote dental health	**Health basis of MAC:**Moderate dental fluorosis (based on cosmetic effect, not health)	Beneficial in preventing dental caries
DBP	Formaldehyde (1997)	None required		By-product of disinfection with ozone; releases from industrial effluents		Guideline value not necessary, as levels in drinking water are below the level at which adverse health effects may occur

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
O	Gasoline and its organic constituents (1986, 2005)	None required		Spill or leaking storage tank		No MAC due to complex composition of gasoline; strong taste and odour at concentrations well below those potentially eliciting adverse health effects (see benzene, ethylbenzene, toluene and xylenes for more information)
P	Glyphosate (1987, 2005)	0.28		Leaching and/or runoff from various uses in weed control	**Health basis of MAC:**Reduced body weight gain	Not expected to migrate to groundwater
DBP	Haloacetic acids - Total (HAAs)Table 2 footnote3(2008)	0.08		By-product of drinking water disinfection with chlorine	**Health basis of MAC:** Liver cancer (DCA); DCA is classified as probably carcinogenic to humans	Refers to the total of monochloroacetic acid (MCA), dichloroacetic acid (DCA), trichloroacetic acid (TCA), monobromoacetic acid (MBA) and dibromoacetic acid (DBA); MAC is based on ability to achieve HAA levels in distribution systems without compromising disinfection; precursor removal limits formation
		ALARA			**Other:** Other organ cancers (DCA, DBA, TCA); liver and other organ effects (body, kidney and testes weights) (MCA)	
T	Hardness (1979)	None required		Naturally occurring (sedimentary rock erosion and seepage, runoff from soils); levels generally higher in groundwater	Although hardness may have significant aesthetic effects, a guideline has not been established because public acceptance of hardness may vary considerably according to the local conditions; major contributors to hardness -- calcium and magnesium -- are not of direct public health concern	Hardness levels between 80 and 100 mg/L (as $CaCO_3$) provide acceptable balance between corrosion and incrustation; where a water softener is used, a separate unsoftened supply for cooking and drinking purposes is recommended

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
I	Iron (1978, 2005)		AO: ≤ 0.3	Naturally occurring (erosion and weathering of rocks and minerals); acidic mine water drainage, landfill leachates, sewage effluents and iron-related industries		Based on taste and staining of laundry and plumbing fixtures; no evidence exists of dietary iron toxicity in the general population
I	Lead (1992)	0.01		Leaching from plumbing (pipes, solder, brass fittings and lead service lines)	**Health basis of MAC:**Biochemical and neurobehavioural effects (intellectual development, behaviour) in infants and young children (under 6 years) **Other:** Anaemia, central nervous system effects; in pregnant women, can affect the unborn child; in infants and children under 6 years, can affect intellectual development, behaviour, size and hearing; classified as probably carcinogenic to humans	Because the MAC is based on chronic effects, it is intended to apply to average concentrations in water consumed for extended periods. Exposure to lead should nevertheless be kept to a minimum; plumbing should be thoroughly flushed before water is used for consumption; most significant contribution is generally from lead service line entering the building
I	Magnesium (1978)	None required		Naturally occurring (erosion and weathering of rocks and minerals)		Guideline value not necessary, as there is no evidence of adverse health effects from magnesium in drinking water
P	Malathion (1986, 2005)	0.19		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	Not expected to leach into groundwater
I	Manganese (1987)		AO: ≤ 0.05	Naturally occurring (erosion and weathering of rocks and minerals)		Based on taste and staining of laundry and plumbing fixtures
I	Mercury (1986)	0.001		Releases or spills from industrial effluents; waste disposal; irrigation or drainage of areas where agricultural pesticides are used	**Health basis of MAC:**Irreversible neurological symptoms	Applies to all forms of mercury; mercury generally not found in drinking water, as it binds to sediments and soil

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
P	2-Methyl-4-chlorophenoxyacetic acid (MCPA) (2010)	0.1		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:**Kidney effects (increased absolute and relative weights, urinary bilirubin, crystals and pH) **Other:** Systemic, liver, testicular, reproductive/developmental and nervous system effects	Can potentially leach into groundwater
O	Methyl tertiary-butyl ether (MTBE) (2006)		AO: ≤ 0.015	Spills from gasoline refineries, filling stations and gasoline-powered boats; seepage into groundwater from leaking storage tanks	There exist too many uncertainties and limitations in the MTBE database to develop a health based guideline.	AO based on odour; levels above the AO would render water unpalatable; as the AO is lower than levels associated with potential toxicological effects, it is considered protective of human health.
P	Metolachlor (1986)	0.05		Leaching and/or runoff from agricultural or other uses	**Health basis of MAC:** Liver lesions and nasal cavity tumours	Readily binds to organic matter in soil; little leaching expected in soils with high organic and clay content
P	Metribuzin (1986, 2005)	0.08		Leaching and/or runoff from agricultural use	**Health basis of MAC:** Liver effects (increased incidence and severity of mucopolysaccharide droplets)	Leaching into groundwater depends on the organic matter content of the soil
O	Monochlorobenzene (1987)	0.08	AO: ≤ 0.03	Releases or spills from industrial effluents	**Health basis of MAC:**Reduced survival and body weight gain	AO based on odour; levels above the AO would render water unpalatable
I	Nitrate/nitrite (1987)	*Nitrate*: 45 as nitrate; 10 as nitrate-nitrogen	*Nitrite (if measured separately)*: 3.2 as nitrite; 1.0 as nitrite-nitrogen	Naturally occurring; leaching or runoff from agricultural fertilizer use, manure and domestic sewage; may be produced from excess ammonia or from microbial activity in distribution systems	**Health basis of MAC:**Methaemoglobinaemia (blue baby syndrome) in infants less than 3 months old (short term) **Other:** Classified as possible carcinogen	MACs are protective of children and adults; systems using chloramine disinfection or that have naturally occurring ammonia should monitor nitrite and nitrate in distribution system
I	Nitrilotriacetic acid (NTA) (1990)	0.4		Sewage contamination	**Health basis of MAC:**Kidney effects (nephritis and nephrosis)	

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
					Other: Classified as possible carcinogen	
DBP	*N*-Nitroso dimethylamine (NDMA) (2010)	0.000 04		By-product of drinking water disinfection with chlorine or chloramines; industrial and sewage treatment plant effluents	**Health basis of MAC:** Liver cancer (classified as probable carcinogen)	MAC considers additional exposure through showering and bathing; levels should be kept low by preventing formation during treatment
A	Odour (1979, 2005)		Inoffensive	Biological or industrial sources		Important to provide drinking water with no offensive odour, as consumers may seek alternative sources that are less safe
P	Paraquat (1986, 2005)	0.01 as paraquat dichloride; 0.007 as paraquat ion		Leaching and/or runoff from agricultural and other uses; added directly to water to control aquatic weeds	**Health basis of MAC:**Various effects on body weight, spleen, testes, liver, lungs, kidney, thyroid, heart and adrenal gland	Entry into drinking water unlikely from crop applications (clay binding); however, may persist in water for several days if directly applied to water
O	Pentachlorophenol (1987, 2005)	0.06	AO: ≤ 0.03	By-product of drinking water disinfection with chlorine; industrial effluents	**Health basis of MAC:**Reduced body weight, changes in clinical parameters, histological changes in kidney and liver, reproductive effects (decreased neonatal survival and growth)	AO based on odour; levels above the AO would render drinking water unpalatable
T	pH (1979)		6.5-8.5Table 2 footnote4	Not applicable		pH can influence the formation of disinfection by-products and effectiveness of treatment
P	Phorate (1986, 2005)	0.002		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	Some potential to leach into groundwater
P	Picloram (1988, 2005)	0.19		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:**Changes in body and liver weights and clinical chemistry parameters **Other:** Kidney effects (liver to body weight ratios and histopathology)	Significant potential to leach into groundwater

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
I	Selenium (1992)	0.01		Naturally occurring (erosion and weathering of rocks and soils)	**Health basis of MAC:**Essential nutritional element **Other:** Hair loss and weakened nails at extremely high levels of exposure	Most exposure from food; little information on toxicity of selenium from drinking water
I	Silver (1986, 2005)	None required		Naturally occurring (erosion and weathering of rocks and soils)		Guideline value not required as drinking water contributes negligibly to an individual's daily intake
P	Simazine (1986)	0.01		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:** Body weight changes and effects on serum and thyroid gland	Extent of leaching decreases with increasing organic matter and clay content
I	Sodium (1979)		AO: ≤ 200	Naturally occurring (erosion and weathering of salt deposits and contact with igneous rock, seawater intrusion); sewage and industrial effluents; sodium-based water softeners		Based on taste; where a sodium-based water softener is used, a separate unsoftened supply for cooking and drinking purposes is recommended
I	Sulphate (1994)		AO: ≤ 500	Industrial wastes	High levels (above 500 mg/L) can cause physiological effects such as diarrhoea or dehydration	Based on taste; health authorities should be notified of drinking water sources containing above 500 mg/L
I	Sulphide (1992)		AO: ≤ 0.05	Can occur in the distribution system from the reduction of sulphates by sulphate-reducing bacteria; industrial wastes		Based on taste and odour; levels above the AO would render water unpalatable
A	Taste (1979, 2005)		Inoffensive	Biological or industrial sources		Important to provide drinking water with no offensive taste, as consumers may seek alternative sources that are less safe

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
T	Temperature (1979, 2005)		AO: ≤ 15°C	Not applicable		Temperature indirectly affects health and aesthetics through impacts on disinfection, corrosion control and formation of biofilms in the distribution system
P	Terbufos (1987, 2005)	0.001		Leaching and/or runoff from agricultural and other uses	**Health basis of MAC:**Nervous system effects (cholinesterase inhibition)	Based on analytical achievability
O	Tetrachloroethylene (1995)	0.03		Industrial effluents or spills	**Health basis of MAC:**Increased liver and kidney weights **Other:** Classified as possible carcinogen; limited evidence of an increased risk of spontaneous abortion	Readily leaches into groundwater; MAC considers additional exposure through showering and bathing
O	2,3,4,6-Tetrachlorophenol (1986, 2005)	0.1	AO: ≤ 0.001	By-product of drinking water disinfection with chlorine; industrial effluents and use of pesticides	**Health basis of MAC:**Developmental effects (embryotoxicity)	AO based on odour; levels above the AO would render drinking water unpalatable
O	Toluene (1986, 2005)		AO: ≤ 0.024	Release of effluents or spills from petroleum and chemical industries		AO based on odour; levels above the AO would render drinking water unpalatable
A	Total dissolved solids (TDS) (1991)		AO: ≤ 500	Naturally occurring; sewage, urban and agricultural runoff, industrial wastewater		Based on taste; TDS above 500 mg/L results in excessive scaling in water pipes, water heaters, boilers and appliances; TDS is composed of calcium, magnesium, sodium, potassium, carbonate, bicarbonate, chloride, sulphate and nitrate
O	Trichloroethylene (2005)	0.005		Industrial effluents and spills from improper disposal	**Health basis of MAC:**Developmental effects (heart malformations) **Other:** Classified as probable carcinogen	MAC considers additional exposure through showering and bathing
O	2,4,6-Trichlorophenol (1987, 2005)	0.005	AO: ≤ 0.002	By-product of drinking water disinfection with chlorine; industrial	**Health basis of MAC:** Liver cancer (classified as probable carcinogen)	AO based on odour; levels above the AO would render drinking water unpalatable

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Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
				effluents and spills		
P	Trifluralin (1989, 2005)	0.045		Runoff from agricultural uses	**Health basis of MAC:**Changes in liver and spleen weights and in serum chemistry	Unlikely to leach into groundwater
DBP	TrihalomethanesTable 2 footnote3(THMs) (2006)	0.1		By-product of drinking water disinfection with chlorine; industrial effluents	**Health basis of MAC:** Liver effects (fatty cysts) (chloroform classified as possible carcinogen) **Other:** Kidney and colorectal cancers	Considers the most commonly found THMs, namely chlorodibromomethane, chloroform, bromodichloromethane and bromoform; MAC based on health effects of chloroform and considers additional exposure through showering and bathing; precursor removal limits formation
I	Uranium (1999)	0.02		Naturally occurring (erosion and weathering of rocks and soils); mill tailings; emissions from nuclear industry and combustion of coal and other fuels; phosphate fertilizers	**Health basis of MAC:**Kidney effects (various lesions); may be rapidly reversible after exposure ceases	Based on treatment achievability; MAC based on chemical effects, as uranium is only weakly radioactive; uranium is rapidly eliminated from the body
O	Vinyl chloride (1992)	0.002		Industrial effluents; degradation product from trichloroethylene and tetrachloroethylene in groundwater; leaching from polyvinyl chloride pipes	**Health basis of MAC:** Liver cancer (classified as human carcinogen) **Other:** Raynaud's disease, effects on bone, circulatory system, thyroid, spleen, central nervous system	Based on treatment and analytical achievability; leaching from polyvinyl chloride pipe is not expected to be significant
O	Xylene (1986, 2005)		AO: ≤ 0.3	Industrial effluents and spills		AO based on taste and odour; levels above the AO would render water unpalatable

Guidelines for Canadian Drinking Water Quality (Aug 2012)

Table 2 - Chemical and Physical Properties

Type	Parameter approval, OR reaffirmation	MAC (mg/IL	Other value (mg/L)	Common sources of parameter in water	Health considerations	Comments
I	Zinc (1979, 2005)		AO: ≤ 5.0	Naturally occurring; industrial and domestic emissions; leaching may occur from galvanized pipes, hot water tanks and brass fittings		AO based on taste; water with zinc levels above the AO tends to be opalescent and develops a greasy film when boiled; plumbing should be thoroughly flushed before water is consumed

Appendix B – Water Quality Test Results – 2013:

Holland Lake

	Coliforms							Colour						
Date	total	E-coli	Alkalinity	Calcium	Bromide	TDS	PH	TCU	ACU	Turbidity	DOC	TOC	UVT	THM
17-Jan			4.63	1.35	<.01		6.2	12	38	1.9				
31-Jan								14	54	2.3				
07-Feb	48	<1						13	100	4.8			56.7	
14-Feb			5.9	1.68			6.2	12	88	4.6				
21-Feb	59	<1				25		14	100	5.2			53.8	
28-Feb								14	112	5.4				
07-Mar	51	<1						12	100	5.5			51	
13-Mar			6.5	1.78	<.01		6.3	8	120	5.2				
21-Mar	39	<1				16		10	64	3.1	2.54	2.9	60.4	
26-Mar								10	47	2.4				360
04-Apr	36	<1						13	37	1.2			75.7	
11-Apr			2.5	0.954	<.01		6.4	11	18	0.6				
18-Apr	190	<1				27		10	12	0.7	2.09	2.22	81.2	
25-Apr								10	12	0.9				
02-May	730	<1						14	27	0.8			82.1	
09-May			2.4	1.06	<.01		6.4	11	19	0.7				
16-May	870	<1						12	16	0.8	2.57	3.05	82.7	
22-May								10	14	0.8				
30-May	230	<1						5	10	0.7			81.5	
06-Jun			2.8	1.15	<.01		6.2	7	10	0.6				
13-Jun	550	<1				19		10	15	0.6	2.58	2.62	82.1	
20-Jun								10	15	0.8				
27-Jun								15	15	0.7			78	350
03-Jul			2.8	1.2	<.01		6.2	10	15	0.8				
11-Jul	550	<1				13		5						
17-Jul								5	10	1				
01-Aug			3.3	1.26	<.01		6.5	10	10	1				
08-Aug	690	1						5	13	0.7	2.48		82.2	
14-Aug								10	13	0.3		2.81		
22-Aug	340	<1						12	14	0.6			82.4	
28-Aug			3.4	1.26	<.01		6.6	12	17	0.6				
05-Sep	730	2				24		9	13	0.7	2.66	3.04	83.2	
12-Sep								13	19	0.9				270
25-Sep			3	1.42			6.6	11	16	0.6				
03-Oct	310	15				22		9	15	0.7	2.98	4.01	81.7	
09-Oct								10	12	0.7				
17-Oct	110	1						10	15	1.2			82.6	
23-Oct			2.8	1.17	<.01		6.5	10	10	0.7				
30-Oct						23		5	11	0.9	2.31	3.48	84.1	
07-Nov	210	1						9	11	0.8				
13-Nov	360	<1						12	11	0.7			83.1	
28-Nov	73	1				13		10	11	0.7	2.73	3.06	82.5	
05-Dec			3	1.4	<.01		6.5	10	14	0.7				
12-Dec								10	12	0.7			85.2	
19-Dec						14		12	17	0.7				
AVE	**325.05**		**3.59**	**1.31**		**19.60**	**6.38**	**10.36**	**30.05**	**1.50**	**2.55**	**3.02**	**76.61**	**326.7**

Stocking Lake Lab Results, 2013

	Coliforms							Colour						
Date	total	E-coli	Alkalinity	Calcium	Bromide	TDS	PH	TCU	ACU	Turbidity	DOC	TOC	UVT	THM
03-Jan	75	<1						10	12	1.2			84.8	
10-Jan								10	13	0.3				
17-Jan			8.5	3.17	<.01		6.9	11	15	0.5				
24-Jan	16	<1				26		14	18	0.4	2.19	2.79	82.3	
31-Jan								12	24	0.5				
07-Feb	13	<1						14	17	0.3			83.5	
14-Feb			8.2	3.21			6.7	12	14	0.5				
21-Feb	5	<1				25		13	18	0.7	1.92	2.17	78.5	
28-Feb								13	17	0.4				
07-Mar	14	<1						13	16	0.4			82.7	
13-Mar			8.5	3.25	<.01		6.9	12	17	0.3				
21-Mar	10	<1				24		10	12	0.5	2.39	2.64	83.3	
27-Mar								12	15	0.5				340
04-Apr	11	<1						10	13	1.1			83.1	
11-Apr			8	3.03	<.01		7	11	14	0.3				
18-Apr	20	<1				40		5	10	0.4	1.82	1.99	83.3	
25-Apr								10	12	0.2				
02-May	11	<1						14	19	0.3			83.1	
08-May			7.9	3.5	<.01		7	13	15	0.3				
16-May	66	<1				28		11	14	0.3	3.13	3.15	84	
22-May								10	13	0.5				
30-May	83	1						5	5	0.6			84.2	
06-Jun			9	3.59	<.01		7	10	13	0.3				
13-Jun	52	<1				42		5	5	0.3	2.39	2.54	87.8	
20-Jun								10	10	0.4				
27-Jun								10	10	0.5			83.9	320
04-Jul				3.59	<.01		7	5	5	0.4				
11-Jul	190	2				25		5	10	1			84.7	
18-Jul								5	10	0.6				
25-Jul	330	<1						10	12	0.4			84.3	
01-Aug			9	4.07	<.01		7.1	10	13	0.5				
08-Aug	730	1						10	13	0.4	2.33		85.8	
15-Aug								5	13	0.3		2.77		
22-Aug	980	2						10	13	0.3			85.7	
28-Aug			8.9	4.24	<.01		6.9	10	16	0.6				
05-Sep	2400	2				30		10	14	0.5	2.74	2.81	85.1	
12-Sep								11	17	0.6				300
19-Sep	2000	1						10	13	0.6			86.2	
26-Sep			9.1	4.56	<.01		7.1	10	15	0.4				
03-Oct	730	3				50		10	15	0.5	2.47	3.92	86.7	
10-Oct								5	5	0.3				
17-Oct	140	1						8	13	0.5			86.5	
24-Oct			10.6	4.73	<0.01		7.1	10	10	0.2				
31-Oct	28	1				27		7	10	0.3	2.27	1.93	88.5	
07-Nov								6	8	0.5				
13-Nov	59	1						10	10	0.5			84.8	
28-Nov	15	<1				27		10	10	0.2	1.65	2.27	88.7	
05-Dec			9.4	1.33	<0.01		7	5	10	0.5				
12-Dec								10	13	0.4			86.7	
19-Dec						13		10	12	0.5				
AVE	**346.87**	**1.50**	**8.83**	**3.52**		**29.75**	**6.98**	**9.64**	**12.82**	**0.46**	**2.30**	**2.63**	**84.73**	**320.00**

Chicken Ladder (Lower Holland Creek) Lab Results, 2013

	Coliforms							Colour						
Date	total	E-coli	Alkalinity	Calcium	Bromide	TDS	PH	TCU	ACU	Turbidity	DOC	TOC	UVT	THM
01-Jan	140	<1						10	13	0.4			84.4	
10-Jan								10	15	0.3				
17-Jan			5.14	1.76	<.01		6.8	14	15	0.1				
24-Jan	47	3				24		16	17	0.2	2.54	2.97	81.3	
31-Jan								14	14	0.4				
07-Feb	91	4						19	26	0.4			71.7	
14-Feb			4.2	1.66	<.01		6.7	13	15	0.2				
21-Feb	70	<1				29		14	17	0.2	2.08	2.12	80.5	
28-Feb								14	18	0.2				
07-Mar	93	<1						13	17				79.7	
13-Mar			3.4	1.19	1.36		6.7	15	20	0.6				
21-Mar	55	3				18		12	15	1.9	2.73	2.84	74.3	
27-Mar								14	15	0.3				360
04-Apr	36	<1						15	17	1.1			79.5	
11-Apr			3.3	1.4	<.01		6.7	20	28	0.6				
18-Apr	99	13				39		10	12	0.2	1.14	1.78	81.8	
25-Apr								10	15	0.4				
02-May	99	<1						12	19	0.3			82.8	
09-May			4.1	1.65	<.01		6.7	13	16	0.3				
16-May	550	3				31		13	15	0.4	2.61	2.76	82.1	
22-May								10	14	0.8				
30-May	520	3						13	13	0.8			70.4	
06-Jun			6.3	2.04	<0.01		6.9	10	11	0.1				
13-Jun	770	7				37		10	15	0.3	2.15	2.35	82.2	
20-Jun								10	10	0.2				
27-Jun								20	25	0.4			72.6	520
04-Jul				2.14	<0.01		6.8	10	10	1				
11-Jul	1600	4				30		8	12	0.2				
17-Jul								10	10	0.3				
25-Jul	1700	6						10	12	0.3			85.2	
01-Aug			5	1.98	<.01		6.8	5	10	0.3				
08-Aug	1200	<1						8	11	0.9	1.9		86.5	
15-Aug								15	15	0.1		2.6		
22-Aug	1700	1						7	10	0.1			87.5	
29-Aug			5.1	2.16	<.01		6.6	11	16	0.2				
05-Sep	1300	4				23		13	14	0.1	1.98	2.07	87.1	
12-Sep								10	12	0.2				190
19-Sep	1700	2						10	14	0.3			88.1	
26-Sep			4.7	2.03	<.01		6.8	9	13	0.3				
03-Oct	1300	12				62		20	25	0.6	5.34	5.65	64.3	
10-Oct								13	13	0.4				
17-Oct	610	1						12	12	0.8			80.7	
24-Oct			6.6	2.56	<0.01		6.9	15	16	0.3				
31-Oct	690	21				23		11	14	0.2	2	1.88	85.7	
07-Nov								25	40	2.4				
13-Nov	520	3						30	30	0.2			68.7	
28-Nov	110	<1				29		12	13	<.1	2.58	2.79	80.6	
05-Dec			6.2	2.73	<.01		6.8	10	12	0.2				
12-Dec								10	12	0.2			82.8	
19-Dec						21		10	15	0.6				
AVE	**652.17**	**5.63**	**4.91**	**1.94**		**30.50**	**6.77**	**12.76**	**15.76**	**0.44**	**2.46**	**2.71**	**80.02**	**356.67**

Appendix C – Town of Ladysmith Long Term Capital Plan (at end of 2013)

ID	Title	Description	Purpose	Impact if Not proceeding	Priority	Year	Cost
53	Chlorination Facility	Construction of a new Chlorination Facility	To upgrade the present chlorination facility.	Will not meet Health Ministry permit requirements.	Committed	2013	$1,093,000
48	Gate Improvements - South Watts Rd	Upgrading of the existing gate at South Watts Rd to improve access control in the south watershed.	To improve access management into the South Watts area.		High	2013	$25,000
13	Holland Dam Safety Improvements	Construct a high flow emergency spillway, other improvements required by BC Dam Safety branch.	To comply with BC Dam Safety Inspection requirements.		High	2013	$10,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2013	$5,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2013	$5,000
8	Hydro Power Generation: Lower Stocking Main	Energy recovery of Water Supply Pipeline on the Lower section of the Stocking lake Supply main.	To generate revenues from the recover of energy from our supply main.		Low	2013	$910,000
14	Water Quality Monitoring Equipment: Holland, Stocking Lakes and Chicken Ladder	Installation of in-situ Water Quality Monitoring Equipment and telemetry to allow for the ongoing management of our water supplies, as well as water modeling software for our system.	To allow us to have ongoing accurate water quality data year round at all sources irrespective of whether the source is being used at the time.		High	2014	$90,000
70	Watershed Modelling Update	Update of previous watershed models completed in the 1990's. Will include impacts from climate change. Growth Targets will be 18k, and 30k.	To verify long term water supply capacity to serve the Town to beyond 2050 at current growth rates.	Possible water supply issues in the long term future.	Committed	2014	$35,000
12	Stocking Lake Dam Repair	Repair of Stocking lake Dam (50/50 split with CVRD Water).	To eliminate leaks in current dam face, and repair the spillway.		High	2014	$200,000
53	Chlorination Facility	Construction of a new Chlorination Facility	To upgrade the present chlorination facility.	Will not meet Health Ministry permit requirements.	Committed	2014	$507,240
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2014	$15,000
85	Watershed Security	Various Initiatives to manage security in the Holland and Stocking lake Watersheds.			High	2014	$50,000

ID	Title	Description	Purpose	Impact if Not proceeding	Priority	Year	Cost
86	Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.		High	2014	$100,000
72	Water Filtration Pilot Project	Temporary Small Scale Pilot Treatment Plant, located next to new Chlorination Building.	To determine the appropriate technology to use for Water Filtration.	Possible Order from VIHA to proceed with Filtration.	High	2014	$165,000
73	Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	Possible VIHA Order.	High	2015	$700,000
12	Stocking Lake Dam Repair	Repair of Stocking lake Dam (50/50 split with CVRD Water).	To eliminate leaks in current dam face, and repair the spillway.		High	2015	$200,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2015	$15,000
86	Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.		High	2015	$300,000
15	Stocking Lake Supply Main Replacement - Phase I	Replacement of the existing AC watermain from Stocking lake to the balancing reservoir.	To provide a more secure Supply main.	Possible loss of water supply from Stocking Lake (and Holland in future)	Medium	2015	$800,000
73	Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	Possible VIHA Order.	High	2016	$1,300,000
74	Holland to Stocking Supply Main	New Supply main to connect Stocking Supply Main with Holland Lake.	To provide a direct system connection from Holland Lake to the Town's water system.		High	2016	$1,500,000
73	Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	Possible VIHA Order.	High	2016	$2,700,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2016	$15,000
86	Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.		High	2016	$300,000
74	Holland to Stocking Supply Main	New Supply main to connect Stocking Supply Main with Holland Lake.	To provide a direct system connection from Holland Lake to the Town's water system.		High	2016	$2,500,000
17	Stocking Lake Supply Main Replacement - Phase II	Replacement of the AC Supply watermain from balancing reservoir to the Old Chlorination Station.		Possible loss of water supply from Stocking Lake (and Holland in future)	Medium	2016	$800,000

ID	Title	Description	Purpose	Impact if Not proceeding	Priority	Year	Cost
73	Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	Possible VIHA Order.	High	2017	$1,300,000
73	Water Filtration Project	Design and Construction for new Water Filtration Plant near Arbutus Reservoir.	To provide Filtration as per VIHA 4-3-2-1 Water Supply Rule.	Possible VIHA Order.	High	2017	$2,700,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2017	$15,000
75	Chicken Ladder Intake Upgrade	Replacement and Upgrade of Chicken Ladder Intake (Lower Holland Creek).	To improve reliability and safety of water supply.		High	2017	$500,000
75	Chicken Ladder Intake Upgrade	Replacement and Upgrade of Chicken Ladder Intake (Lower Holland Creek).	To improve reliability and safety of water supply.		High	2017	$1,500,000
76	Chicken Ladder Supply Main replacement	Replacement of 1960's AC supply main from Chicken Ladder to Water Treatment Plant.			High	2017	$1,000,000
86	Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.		High	2017	$300,000
79	Energy Recovery Plant - Old PRV Station				Medium	2017	$550,000
79	Energy Recovery Plant - Old PRV Station				Medium	2017	$100,000
77	Replace Holland Supply Main: Pub Works Yard to Colonia	Replacement of old AC supply main along the Holland Creek Trail			Medium	2017	$330,000
86	Watermain Replacement Program	Annual Capital watermain replacement Program in the Town's Distribution System.	To replace aging AC and Cast Iron mains throughout the distribution system.		High	2018	$300,000
11	Low Flush Toilet Program	Continuation of the Town's Low Flush Toilet Program.	To reduce water consumption in the Town.	Possible reduction in number of low flush toilets installed.	High	2018	$15,000
82	Upper Pressure Zone - Pump Station				Medium	2018	$650,000
82	Upper Pressure Zone - Pump Station				Medium	2018	$20,000
83	Upper pressure Zone - Supply Main	To connect lower pressure zone with new reservoir.			Medium	2018	$300,000
84	Arbutus reservoir - Capacity Increase	Doubling of Arbutus Reservoir			Medium	2018	$700,000
80	Energy Recovery Plant - Stocking Lake				Medium	2018	$20,000
84	Arbutus reservoir - Capacity Increase	Doubling of Arbutus Reservoir			Medium	2018	$20,000

ID	Title	Description	Purpose	Impact if Not proceeding	Priority	Year	Cost
83	Upper pressure Zone - Supply Main	To connect lower pressure zone with new reservoir.			Medium	2018	$15,000
80	Energy Recovery Plant - Stocking Lake				Medium	2018	$550,000
81	New Upper Pressure Zone Reservoir	To service Couverdon, possibly others.			Medium	2018	$20,000
78	Replace Supply Main: Colonia to Treatment Plant				Medium	2018	$200,000
81	New Upper Pressure Zone Reservoir	To service Couverdon, possibly others.			Medium	2018	$500,000

Appendix D – Additional Information - Haloacetic Acids in Domestic water Supplies

The following additional information is contained within the report referenced at:

7.0 Treatment Technology

Although HAA formation in water is largely a function of the amount of organic compounds in water and their contact time with chlorine, it is important to recognize that the use of chlorination and other disinfection processes has virtually eliminated waterborne microbial diseases. In order to reduce HAA levels in the finished water, it is important to characterize the source water to ensure that the treatment process is optimized for precursor removal.

7.1 Municipal scale

There are three approaches to limiting the concentrations of HAAs in municipally treated drinking water:

- treatment of water to remove HAA precursors prior to disinfection;
- the use of alternative disinfectants and disinfection strategies; and
- treatment of water to remove HAAs after their formation.

The majority of changes occurring in the water industry now focus on strategies to remove DBP precursors prior to disinfection and the use of alternative disinfectants and alternative disinfection strategies.

7.1.1 Removal of precursors prior to municipal disinfection

The removal of organic precursors is the most effective way to reduce the concentrations of all DBPs, including HAAs, in finished water (U.S. EPA, 1999c; Reid Crowther & Partners Ltd., 2000). These precursors include synthetic organic compounds and NOM, which can react with disinfectants to form HAAs. Removing HAA precursors will also result in the formation of lower concentrations of HAAs (Reid Crowther & Partners Ltd., 2000). Conventional municipal-scale water treatment techniques (coagulation, sedimentation, dissolved air flotation, precipitative softening and filtration) can reduce the amount of HAA precursors, but are ineffective in removing HAAs once they are formed. Granular activated carbon (GAC), membranes and ozone-biofiltration systems can also remove organic matter from water. The U.S. EPA has identified precursor removal technologies such as GAC and membrane filtration, specifically nanofiltration, as Best Available Technologies (BAT) for controlling DBP formation (U.S. EPA, 2005b).

Potassium permanganate can be used to oxidize organic precursors at the head of the treatment plant, thus minimizing the formation of by-products at the disinfection stage (U.S. EPA, 1999a). The use of ozone for oxidation of precursors is currently being studied. Early work has shown that the effects of ozonation, prior to chlorination, depend on treatment design and raw water quality and thus are unpredictable. The key variables that seem to determine the effect of ozone are dose, pH, alkalinity and the nature of the organic material in the water. Ozone has been shown to be effective at reducing precursors at low pH. However, at pH levels above 7.5, ozone may actually increase the production of CDBP precursors (U.S. EPA, 1999a).

7.1.2 Alternative municipal disinfection strategies

The use of alternative disinfectants, such as chloramines (secondary disinfection only), ozone (primary disinfection only) and chlorine dioxide (primary and secondary disinfection), is increasing. However, each of these alternatives has also been shown to form its own set of DBPs. Combinations of disinfectants, when optimized, can help controlHAA formation. Preozonation is feasible for water sources that have turbidity levels below 10 nephelometric turbidity units (NTU) and bromide concentrations below 0.01 mg/L, to minimize the formation of bromate (Reid Crowther & Partners Ltd., 2000). Ultraviolet (UV) disinfection is also being used as an alternative disinfectant. Since UV disinfection is dependent on light transmission to the microbes, water quality characteristics affectingUV transmittance must be considered in the design of the system. UV irradiation at typical doses and wavelengths does not affect HAA formation in subsequent chlorination or chloramination steps (Reid Crowther & Partners Ltd., 2000). Neither ozone nor UVdisinfection leaves a residual disinfectant, and both must therefore be used in combination with a secondary disinfectant to maintain a residual in the distribution system.

It is recommended that any change made to the treatment process, particularly when changing the disinfectant, be accompanied by close monitoring of lead levels in the distributed water. A change of disinfectant has been found to affect the levels of lead at the tap; in Washington, DC, for example, a change from chlorine to chloramines resulted in significantly increased levels of lead in the distributed drinking water. When chlorine, a powerful oxidant, is used as the disinfectant, lead dioxide scales formed in distribution system pipes reach a dynamic equilibrium in the distribution system. In Washington, DC, switching from chlorine to chloramines decreased the oxidation-reduction potential of the distributed water and destabilized the lead dioxide scales, which resulted in increased lead leaching (Schock and Giani, 2004). Subsequent laboratory experiments by Edwards and Dudi (2004) and Lytle and Schock (2005) confirmed that lead dioxide deposits could be readily formed and subsequently destabilized in weeks to months under realistic conditions of distribution system pH, oxidation-reduction potential and alkalinity.

7.1.3 Removal of HAAs after formation

Although precursor removal is considered the most effective approach to reduce HAAconcentrations, removal of HAAs is also possible. Adsorption onto activated carbon is widely used to remove organic compounds such as HAAs from drinking water. This method involves pumping water through a bed of activated carbon onto which HAA molecules become attached (adsorbed). If an activated carbon filter bed is deep enough to allow sufficient contact time, it can be effective in removing HAAs from drinking water. Biofiltration may be an effective process for removing biodegradable organic matter and biodegradable DBPs from water. GAC, anthracite, sand and garnet are common media that support colonization by bacteria and can be used as biological filters. Information on the use of biofiltration for HAA removal is limited, although work has shown that bacteria-colonized GAC (biologically active carbon) is an effective process for HAA removal (Xie, 2004).

Appendix E – Ministry of Health – Microbiological Results, 2013

The following pages provide the Ministry of health Microbiological Sampling for The Town's distribution system for 20013.

TOWN OF LADYSMITH WATER WORKS

Facility Location:
410 Esplanade
Ladysmith

Facility Information:
Facility Type: DWT

Facility Sampling History:

Location	Date	Total Coliform	E. Coli
405 Blair Place, 405 Blair Place	19-Feb-2014	L1	L1
432 Davis Road, 432 Davis Road	19-Feb-2014	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	19-Feb-2014	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	12-Feb-2014	L1	L1
558 Hooper Place, 558 Hooper Place	12-Feb-2014	L1	L1
606 OAKWOOD, 606 Oakwood	12-Feb-2014	L1	L1
City Hall , 410 Esplanade	5-Feb-2014	A	
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	5-Feb-2014	A	
1280 Rocky Creek Road, 1280 Rocky Creek Rd	28-Jan-2014	L1	L1
558 Hooper Place, 558 Hooper Place	28-Jan-2014	L1	L1
432 DAVIS ROAD, 432 Davis Road	21-Jan-2014	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	21-Jan-2014	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	21-Jan-2014	L1	L1
604 Farrell Road, 604 Farrell ROad	14-Jan-2014	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	14-Jan-2014	L1	L1
405 Blair Place, 405 Blair Place	7-Jan-2014	L1	L1
606 OAKWOOD, 606 Oakwood	7-Jan-2014	L1	L1
City Hall , 410 Esplanade	7-Jan-2014	L1	L1
432 DAVIS ROAD, 432 Davis Road	16-Dec-2013	L1	L1
City Hall , 410 Esplanade	16-Dec-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	16-Dec-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	9-Dec-2013	L1	L1
558 Hooper Place, 558 Hooper Place	9-Dec-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	9-Dec-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	9-Dec-2013	L1	L1
405 Blair Place, 405 Blair Place	2-Dec-2013	L1	L1
606 OAKWOOD, 606 Oakwood	2-Dec-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	2-Dec-2013	L1	L1

1280 Rocky Creek Road, 1280 Rocky Creek Rd	26-Nov-2013	L1	L1
558 Hooper Place, 558 Hooper Place	26-Nov-2013	L1	L1
City Hall , 410 Esplanade	26-Nov-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	19-Nov-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	19-Nov-2013	L1	L1
405 Blair Place, 405 Blair Place	13-Nov-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	13-Nov-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	5-Nov-2013	L1	L1
606 OAKWOOD, 606 Oakwood	5-Nov-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	5-Nov-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	29-Oct-2013	L1	L1
558 Hooper Place, 558 Hooper Place	29-Oct-2013	L1	L1
City Hall , 410 Esplanade	22-Oct-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	22-Oct-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	15-Oct-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	15-Oct-2013	L1	L1
405 Blair Place, 405 Blair Place	8-Oct-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	8-Oct-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	1-Oct-2013	L1	L1
606 OAKWOOD, 606 Oakwood	1-Oct-2013	L1	L1
558 Hooper Place, 558 Hooper Place	24-Sep-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	24-Sep-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	17-Sep-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	17-Sep-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	17-Sep-2013	L1	L1
405 Blair Place, 405 Blair Place	10-Sep-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	10-Sep-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	4-Sep-2013	L1	L1
606 OAKWOOD, 606 Oakwood	4-Sep-2013	L1	L1
City Hall , 410 Esplanade	4-Sep-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	27-Aug-2013	L1	L1
City Hall , 410 Esplanade	27-Aug-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	27-Aug-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	21-Aug-2013	L1	L1
558 Hooper Place, 558 Hooper Place	21-Aug-2013	L1	L1
606 OAKWOOD, 606 Oakwood	21-Aug-2013	L1	L1
405 Blair Place, 405 Blair Place	13-Aug-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	13-Aug-2013	1	L1
604 Farrell Road, 604 Farrell ROad	6-Aug-2013	L1	L1

Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	6-Aug-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	23-Jul-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	23-Jul-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	16-Jul-2013	L1	L1
558 Hooper Place, 558 Hooper Place	16-Jul-2013	L1	L1
City Hall , 410 Esplanade	16-Jul-2013	L1	L1
405 Blair Place, 405 Blair Place	9-Jul-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	9-Jul-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	3-Jul-2013	L1	L1
606 OAKWOOD, 606 Oakwood	3-Jul-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	3-Jul-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	26-Jun-2013	L1	L1
City Hall , 410 Esplanade	26-Jun-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	26-Jun-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	19-Jun-2013	L1	L1
558 Hooper Place, 558 Hooper Place	19-Jun-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	11-Jun-2013	L1	L1
606 OAKWOOD, 606 Oakwood	11-Jun-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	11-Jun-2013	L1	L1
405 Blair Place, 405 Blair Place	5-Jun-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	5-Jun-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	28-May-2013	L1	L1
558 Hooper Place, 558 Hooper Place	28-May-2013	L1	L1
City Hall , 410 Esplanade	22-May-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	22-May-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	15-May-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	15-May-2013	L1	L1
405 Blair Place, 405 Blair Place	7-May-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	7-May-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	1-May-2013	L1	L1
606 OAKWOOD, 606 Oakwood	1-May-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	16-Apr-2013	L1	L1
City Hall , 410 Esplanade	16-Apr-2013	L1	L1

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◆NEXT

TOWN OF LADYSMITH WATER WORKS

Facility Location:
410 Esplanade
Ladysmith

Facility Information:
Facility Type: DWT

Facility Sampling History:

Location	Date	Total Coliform	E. Coli
Kinsmen Park , 1000 Colonia Drive, Ladysmith	16-Apr-2013	L1	L1
405 Blair Place, 405 Blair Place	9-Apr-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	9-Apr-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	3-Apr-2013	L1	L1
606 OAKWOOD, 606 Oakwood	3-Apr-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	3-Apr-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	26-Mar-2013	L1	L1
558 Hooper Place, 558 Hooper Place	26-Mar-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	26-Mar-2013	L1	L1
AUDIT LADYSMITH, TOWN OF WATERWORKS	19-Mar-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	19-Mar-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	19-Mar-2013	L1	L1
405 Blair Place, 405 Blair Place	12-Mar-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	12-Mar-2013	L1	L1
City Hall , 410 Esplanade	12-Mar-2013	L1	L1
606 OAKWOOD, 606 Oakwood	5-Mar-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	5-Mar-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	26-Feb-2013	L1	L1
558 Hooper Place, 558 Hooper Place	26-Feb-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	19-Feb-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	19-Feb-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	19-Feb-2013	L1	L1
405 Blair Place, 405 Blair Place	13-Feb-2013	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	13-Feb-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	5-Feb-2013	L1	L1
606 OAKWOOD, 606 Oakwood	5-Feb-2013	L1	L1
City Hall , 410 Esplanade	5-Feb-2013	L1	L1
City Hall , 410 Esplanade	29-Jan-2013	L1	L1

Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	29-Jan-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	22-Jan-2013	L1	L1
558 Hooper Place, 558 Hooper Place	22-Jan-2013	L1	L1
432 DAVIS ROAD, 432 Davis Road	15-Jan-2013	L1	L1
606 OAKWOOD, 606 Oakwood	15-Jan-2013	L1	L1
604 Farrell Road, 604 Farrell ROad	9-Jan-2013	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	9-Jan-2013	L1	L1
405 Blair Place, 405 Blair Place	2-Jan-2013	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	2-Jan-2013	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	18-Dec-2012	L1	L1
558 Hooper Place, 558 Hooper Place	18-Dec-2012	L1	L1
432 DAVIS ROAD, 432 Davis Road	11-Dec-2012	L1	L1
604 Farrell Road, 604 Farrell ROad	11-Dec-2012	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	11-Dec-2012	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	11-Dec-2012	L1	L1
405 Blair Place, 405 Blair Place	5-Dec-2012	L1	L1
606 OAKWOOD, 606 Oakwood	5-Dec-2012	L1	L1
City Hall , 410 Esplanade	5-Dec-2012	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	5-Dec-2012	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	27-Nov-2012	L1	L1
558 Hooper Place, 558 Hooper Place	27-Nov-2012	L1	L1
604 Farrell Road, 604 Farrell ROad	20-Nov-2012	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	20-Nov-2012	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	20-Nov-2012	L1	L1
432 DAVIS ROAD, 432 Davis Road	14-Nov-2012	L1	L1
606 OAKWOOD, 606 Oakwood	14-Nov-2012	L1	L1
City Hall , 410 Esplanade	14-Nov-2012	L1	L1
405 Blair Place, 405 Blair Place	6-Nov-2012	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	6-Nov-2012	L1	L1
1280 Rocky Creek Road, 1280 Rocky Creek Rd	30-Oct-2012	L1	L1
Wastewater Treatment Plant, Oyster Cove Road, Ladysmith	30-Oct-2012	L1	L1
604 Farrell Road, 604 Farrell ROad	22-Oct-2012	L1	L1
606 OAKWOOD, 606 Oakwood	22-Oct-2012	L1	L1
405 Blair Place, 405 Blair Place	16-Oct-2012	L1	L1
Public Works Yard, 340 6th Avenue, Ladysmith	16-Oct-2012	L1	L1
558 Hooper Place, 558 Hooper Place	9-Oct-2012	L1	L1
Kinsmen Park , 1000 Colonia Drive, Ladysmith	9-Oct-2012	L1	L1
432 DAVIS ROAD, 432 Davis Road	2-Oct-2012	L1	L1

Appendix F – Lab Results - 2013

The following pages provide the Town's Lab Results for 20013.

Maxxam

Success Through Science

Maxxam Job #: B300549
Report Date: 2013/01/08

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FI0351	FI0352		
Sampling Date		2013/01/03 08:40	2013/01/03 08:00		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS					
Transmittance at 254nm	%T/cm	84.8	84.4	N/A	6465372
Misc. Inorganics					
UV absorbance (254nm)	AU/cm	0.072	0.074	0.010	6470693
MISCELLANEOUS					
Apparent Colour	Col. Unit	12	13	5	6465787
True Colour	Col. Unit	10	10	5	6465786
Physical Properties					
Turbidity	NTU	1.2	0.4	0.1	6465784

MICROBIOLOGY (WATER)

Maxxam ID		FI0351	FI0352		
Sampling Date		2013/01/03 08:40	2013/01/03 08:00		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.					
E. coli	MPN/100mL	<1	<1	1	6465852
Total Coliforms	MPN/100mL	75	140	1	6465852

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B302284
Report Date: 2013/01/11

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FI8959	FI8960		
Sampling Date		2013/01/10 09:00	2013/01/10 08:30		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS					
Apparent Colour	Col. Unit	13	15	5	6481793
True Colour	Col. Unit	10	10	5	6477761
Physical Properties					
Turbidity	NTU	0.3	0.3	0.1	6481084

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B304094
Report Date: 2013/01/24

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FK1649	FK1650	FK1651		
Sampling Date		2013/01/17 07:25	2013/01/17 09:00	2013/01/17 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	6497677
Alkalinity (Total as CaCO3)	mg/L	4.63	8.50	5.14	0.50	6502755
Alkalinity (PP as CaCO3)	mg/L	<0.50	<0.50	<0.50	0.50	6502755
Bicarbonate (HCO3)	mg/L	5.65	10.4	6.27	0.50	6502755
Carbonate (CO3)	mg/L	<0.50	<0.50	<0.50	0.50	6502755
Hydroxide (OH)	mg/L	<0.50	<0.50	<0.50	0.50	6502755
MISCELLANEOUS						
Apparent Colour	Col. Unit	38	15	15	5	6500845
True Colour	Col. Unit	12	11	14	5	6498037
Physical Properties						
pH	pH Units	6.2	6.9	6.8		6494314
Physical Properties						
Turbidity	NTU	1.9	0.5	0.1	0.1	6498193

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		FK1649	FK1650	FK1651		
Sampling Date		2013/01/17 07:25	2013/01/17 09:00	2013/01/17 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1350	3170	1760	50	6500828

RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B306020
Report Date: 2013/01/31

TOWN OF LADYSMITH

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FL6142		FL6143	FL6144		
Sampling Date		2013/01/24 07:15		2013/01/24 08:05	2013/01/24 08:45		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS							
Transmittance at 254nm	%T/cm	70.6	N/A	82.3	81.3	N/A	6515628
Misc. Inorganics							
UV absorbance (254nm)	AU/cm	0.151	0.010	0.084	0.090	0.010	6525285
Dissolved Organic Carbon (C)	mg/L	2.62	0.50	2.19	2.54	0.50	6523635
Total Organic Carbon (C)	mg/L	3.31	0.50	2.79	2.97	0.50	6523637
MISCELLANEOUS							
Apparent Colour	Col. Unit	130	50	18	17	5	6516385
True Colour	Col. Unit	14	5	14	16	5	6516384
Physical Properties							
Total Dissolved Solids	mg/L	18	10	26	24	10	6520410
Turbidity	NTU	6.5	0.1	0.4	0.2	0.1	6516192

MICROBIOLOGY (WATER)

Maxxam ID		FL6142	FL6143	FL6144		
Sampling Date		2013/01/24 07:15	2013/01/24 08:05	2013/01/24 08:45		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	3	1	6519853
Total Coliforms	MPN/100mL	170	16	47	1	6519853

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B307835
Report Date: 2013/02/05

TOWN OF LADYSMITH

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FN1309		FN1310	FN1311		
Sampling Date		2013/01/31 07:20		2013/01/31 08:50	2013/01/31 08:10		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS							
Apparent Colour	Col. Unit	54	10	24	14	5	6534701
True Colour	Col. Unit	14	5	12	14	5	6534697
Physical Properties							
Turbidity	NTU	2.3	0.1	0.5	0.4	0.1	6535288

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B309853
Report Date: 2013/02/13

TOWN OF LADYSMITH

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FO6326	FO6327	FO6328		
Sampling Date		2013/02/07 07:00	2013/02/07 09:05	2013/02/07 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	56.7	83.5	71.7	N/A	6553267
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.246	0.079	0.145	0.010	6558773
MISCELLANEOUS						
Apparent Colour	Col. Unit	100	17	26	5	6553803
True Colour	Col. Unit	13	14	19	5	6553760
Physical Properties						
Turbidity	NTU	4.8	0.3	0.4	0.1	6553545

MICROBIOLOGY (WATER)

Maxxam ID		FO6326	FO6327	FO6328		
Sampling Date		2013/02/07 07:00	2013/02/07 09:05	2013/02/07 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	4	1	6558775
Total Coliforms	MPN/100mL	48	13	91	1	6558775

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B309851
Report Date: 2013/02/14

TOWN OF LADYSMITH

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FO6323		FO6324		
Sampling Date		2013/02/06 15:20		2013/02/06 15:20		
	UNITS	INFLUENT	RDL	EFFLUENT	RDL	QC Batch
Demand Parameters						
Biochemical Oxygen Demand	mg/L	187	5	136	5	6562506
Soluble Biochemical Oxygen Demand	mg/L	86	5	64	5	6562651
Misc. Inorganics						
Alkalinity (Total as CaCO3)	mg/L			141	0.5	6550005
Alkalinity (PP as CaCO3)	mg/L			<0.5	0.5	6550005
Bicarbonate (HCO3)	mg/L			173	0.5	6550005
Carbonate (CO3)	mg/L			<0.5	0.5	6550005
Hydroxide (OH)	mg/L			<0.5	0.5	6550005
Total Suspended Solids	mg/L	237	10	96	5	6553586
Total Suspended Solids (Fixed)	mg/L	<10	10	<5	5	6561816
Total Suspended Solids (Volatile)	mg/L	230	10	94	5	6561816
Nutrients						
Ammonia (N)	mg/L	29	0.050	29	0.050	6558615
Dissolved Phosphorus (P)	mg/L	3.77	0.050	3.48	0.050	6561292
Total Total Kjeldahl Nitrogen (Calc)	mg/L	42.5	1.0	44.5	1.0	6553015
Nitrate plus Nitrite (N)	mg/L	<0.020	0.020	<0.020	0.020	6554492
Total Nitrogen (N)	mg/L	42.5	1.0	44.5	1.0	6559479
Total Phosphorus (P)	mg/L	6.29	0.10	5.59	0.10	6561293
Physical Properties						
Total Solids	mg/L	576	10	406	10	6553830

RDL = Reportable Detection Limit

Maxxam

Success Through Science®

Maxxam Job #: B311767
Report Date: 2013/02/21

TOWN OF LADYSMITH

Your P.O. #: 27327

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FQ0658		FQ0659	FQ0660		
Sampling Date		2013/02/14 06:30		2013/02/14 08:50	2013/02/14 08:05		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics							
Dissolved Bromide (Br)	mg/L	<0.010	0.010	<0.010	<0.010	0.010	6570982
Alkalinity (Total as CaCO3)	mg/L	5.9	0.5	8.2	4.2	0.5	6568250
Alkalinity (PP as CaCO3)	mg/L	<0.5	0.5	<0.5	<0.5	0.5	6568250
Bicarbonate (HCO3)	mg/L	7.2	0.5	10.0	5.1	0.5	6568250
Carbonate (CO3)	mg/L	<0.5	0.5	<0.5	<0.5	0.5	6568250
Hydroxide (OH)	mg/L	<0.5	0.5	<0.5	<0.5	0.5	6568250
MISCELLANEOUS							
Apparent Colour	Col. Unit	88	10	14	15	5	6573388
True Colour	Col. Unit	12	5	12	13	5	6570811
Physical Properties							
pH	pH Units	6.2		6.7	6.7		6565647
Physical Properties							
Turbidity	NTU	4.6	0.1	0.5	0.2	0.1	6569324

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		FQ0658	FQ0659	FQ0660		
Sampling Date		2013/02/14 06:30	2013/02/14 08:50	2013/02/14 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1680	3210	1660	50	6577521

RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B313848
Report Date: 2013/02/27

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FR6059		FR6060	FR6061		
Sampling Date		2013/02/21 08:00		2013/02/21 08:40	2013/02/21 08:00		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS							
Transmittance at 254nm	%T/cm	53.8	N/A	78.5	80.5	N/A	6591054
Misc. Inorganics							
UV absorbance (254nm)	AU/cm	0.270	0.010	0.105	0.094	0.010	6599067
Dissolved Organic Carbon (C)	mg/L	2.62	0.50	1.92	2.08	0.50	6594920
Total Organic Carbon (C)	mg/L	2.78	0.50	2.17	2.12	0.50	6594922
MISCELLANEOUS							
Apparent Colour	Col. Unit	100	50	18	17	5	6591636
True Colour	Col. Unit	14	5	13	14	5	6585928
Physical Properties							
Total Dissolved Solids	mg/L	25	10	25	29	10	6598783
Turbidity	NTU	5.2	0.1	0.7	0.2	0.1	6589097

MICROBIOLOGY (WATER)

Maxxam ID		FR6059	FR6060	FR6061		
Sampling Date		2013/02/21 08:00	2013/02/21 08:40	2013/02/21 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	<1	1	6594794
Total Coliforms	MPN/100mL	59	5	70	1	6594794

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B316044
Report Date: 2013/03/06

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FT0544		FT0545	FT0546		
Sampling Date		2013/02/28 07:00		2013/02/28 08:40	2013/02/28 08:00		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS							
Apparent Colour	Col. Unit	112	20	17	18	5	6612871
True Colour	Col. Unit	14	5	13	14	5	6612869
Physical Properties							
Turbidity	NTU	5.4	0.1	0.4	0.2	0.1	6610391

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		FT0544	FT0545	FT0546		
Sampling Date		2013/02/28 07:00	2013/02/28 08:40	2013/02/28 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	6619337
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	102	102	102		6619337

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B318279
Report Date: 2013/03/14

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FU8561		FU8562	FU8563		
Sampling Date		2013/03/07 07:00		2013/03/07 08:35	2013/03/07 08:00		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS							
Transmittance at 254nm	%T/cm	51.0	N/A	82.7	79.7	N/A	6631671
Misc. Inorganics							
UV absorbance (254nm)	AU/cm	0.292	0.010	0.082	0.099	0.010	6638040
MISCELLANEOUS							
Apparent Colour	Col. Unit	100(1)	20	16	17	5	6636544
True Colour	Col. Unit	12	5	13	13	5	6631522
Physical Properties							
Turbidity	NTU	5.5	0.1	0.4	0.2	0.1	6631518

MICROBIOLOGY (WATER)

Maxxam ID		FU8561	FU8562	FU8563		
Sampling Date		2013/03/07 07:00	2013/03/07 08:35	2013/03/07 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	<1	1	6638315
Total Coliforms	MPN/100mL	51	14	93	1	6638315

N/A = Not Applicable
RDL = Reportable Detection Limit
(1) - Detection limit raised due to dilutions necessary to bring result with calibration range {March 11, 2013 - DN2}.

Maxxam

Success through Service

Maxxam Job #: B320414
Report Date: 2013/03/20

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FW6087	FW6088	FW6089		
Sampling Date		2013/03/13 13:00	2013/03/14 09:05	2013/03/14 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	1.36	0.010	6656274
Alkalinity (Total as CaCO3)	mg/L	6.5	8.5	3.4	0.5	6648368
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	6648368
Bicarbonate (HCO3)	mg/L	7.9	10.3	4.2	0.5	6648368
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	6648368
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	6648368
MISCELLANEOUS						
Apparent Colour	Col. Unit	120	17	20	5	6653171
True Colour	Col. Unit	8	12	15	5	6653161
Physical Properties						
pH	pH Units	6.3	6.9	6.7		6648366
Physical Properties						
Turbidity	NTU	5.2	0.3	0.6	0.1	6653172

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		FW6087	FW6088	FW6089		
Sampling Date		2013/03/13 13:00	2013/03/14 09:05	2013/03/14 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1780	3250	1190	50	6662534

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B322619
Report Date: 2013/03/28

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FY5426		FY5427	FY5428		
Sampling Date		2013/03/21 08:00		2013/03/21 08:50	2013/03/21 08:10		
	UNITS	HOLLAND LAKE	RDL	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS							
Transmittance at 254nm	%T/cm	60.4	N/A	83.3	74.3	N/A	6676491
Misc. Inorganics							
UV absorbance (254nm)	AU/cm	0.219	0.010	0.079	0.129	0.010	6681705
Dissolved Organic Carbon (C)	mg/L	2.54	0.50	2.39	2.73	0.50	6677871
Total Organic Carbon (C)	mg/L	2.90	0.50	2.64	2.84	0.50	6677875
MISCELLANEOUS							
Apparent Colour	Col. Unit	64	10	12	15	5	6677590
True Colour	Col. Unit	10	5	10	12	5	6672868
Physical Properties							
Total Dissolved Solids	mg/L	16	10	24	18	10	6688854
Turbidity	NTU	3.1	0.1	0.5	1.9	0.1	6672870

MICROBIOLOGY (WATER)

Maxxam ID		FY5426	FY5427	FY5428		
Sampling Date		2013/03/21 08:00	2013/03/21 08:50	2013/03/21 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	3	1	6679253
Total Coliforms	MPN/100mL	39	10	55	1	6679253

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B326305
Report Date: 2013/04/10

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GB6654	GB6655	GB6656		
Sampling Date		2013/04/04 07:15	2013/04/04 09:00	2013/04/04 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	75.7	83.1	79.5	N/A	6713599
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.121	0.080	0.100	0.010	6722135
MISCELLANEOUS						
Apparent Colour	Col. Unit	37	13	17	5	6713606
True Colour	Col. Unit	13	10	15	5	6713602
Physical Properties						
Turbidity	NTU	1.2	1.1	1.1	0.1	6713601

MICROBIOLOGY (WATER)

Maxxam ID		GB6654	GB6655	GB6656		
Sampling Date		2013/04/04 07:15	2013/04/04 09:00	2013/04/04 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	<1	1	6715147
Total Coliforms	MPN/100mL	36	11	36	1	6715147

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam Job #: B324176
Report Date: 2013/04/11

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GA2075	GA2076	GA2077		
Sampling Date		2013/03/26 13:45	2013/03/27 09:05	2013/03/27 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	47	15	15	5	6692351
True Colour	Col. Unit	10	12	14	5	6692335
Physical Properties						
Turbidity	NTU	2.4	0.5	0.3	0.1	6692861

VOLATILE ORGANICS BY GC-MS (WATER)

Maxxam ID		GA2075	GA2076	GA2077		
Sampling Date		2013/03/26 13:45	2013/03/27 09:05	2013/03/27 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Volatiles						
Bromodichloromethane	ug/L	6.0	4.9	3.9	3.0	6702062
Bromoform	ug/L	<3.0	<3.0	<3.0	3.0	6702062
Chlorodibromomethane	ug/L	<3.0	<3.0	<3.0	3.0	6702062
Chloroform	ug/L	340	330	350	3.0	6702062
THM Formation Potential	ug/L	360	340	360	3.0	6702062

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B328415
Report Date: 2013/04/17

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GD0626	GD0627	GD0628		
Sampling Date		2013/04/11 08:10	2013/04/11 09:10	2013/04/11 08:30		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	6734157
Alkalinity (Total as CaCO3)	mg/L	2.5	8.0	3.3	0.5	6727401
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	6727401
Bicarbonate (HCO3)	mg/L	3.0	9.8	4.0	0.5	6727401
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	6727401
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	6727401
MISCELLANEOUS						
Apparent Colour	Col. Unit	18	14	28	5	6735720
True Colour	Col. Unit	11	11	20	5	6733808
Physical Properties						
pH	pH Units	6.4	7.0	6.7		6727397
Physical Properties						
Turbidity	NTU	0.6	0.3	0.6	0.1	6733758

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		GD0626	GD0627	GD0628		
Sampling Date		2013/04/11 08:10	2013/04/11 09:10	2013/04/11 08:30		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	954	3030	1400	50	6742058

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B330451
Report Date: 2013/04/24

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GE4954	GE4955	GE4956		
Sampling Date		2013/04/18 08:00	2013/04/18 08:55	2013/04/18 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	81.2	83.3	81.8	N/A	6752248
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.090	0.079	0.087	0.010	6761040
Dissolved Organic Carbon (C)	mg/L	2.09	1.82	1.14	0.50	6753554
Total Organic Carbon (C)	mg/L	2.22	1.99	1.78	0.50	6753555
MISCELLANEOUS						
Apparent Colour	Col. Unit	12	10	12	5	6754025
True Colour	Col. Unit	10	5	10	5	6750022
Physical Properties						
Total Dissolved Solids	mg/L	27	40	39	10	6758520
Turbidity	NTU	0.7	0.4	0.2	0.1	6750025

MICROBIOLOGY (WATER)

Maxxam ID		GE4954	GE4955	GE4956		
Sampling Date		2013/04/18 08:00	2013/04/18 08:55	2013/04/18 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	13	1	6758476
Total Coliforms	MPN/100mL	190	20	99	1	6758476

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam Job #: B332776
Report Date: 2013/04/30

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GF9704	GF9705	GF9706		
Sampling Date		2013/04/25 14:25	2013/04/26 09:15	2013/04/26 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	12	12	15	5	6773676
True Colour	Col. Unit	10	10	10	5	6773695
Physical Properties						
Turbidity	NTU	0.9	0.2	0.4	0.1	6773712

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		GF9704	GF9705	GF9706		
Sampling Date		2013/04/25 14:25	2013/04/26 09:15	2013/04/26 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	6774215
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	102	103	101		6774215

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B334455
Report Date: 2013/05/08

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GH2230	GH2231	GH2232		
Sampling Date		2013/05/02 08:00	2013/05/02 07:15	2013/05/02 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.1	83.1	82.8	N/A	6786527
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.086	0.081	0.082	0.010	6794947
MISCELLANEOUS						
Apparent Colour	Col. Unit	27	19	19	5	6786710
True Colour	Col. Unit	14	14	12	5	6784320
Physical Properties						
Turbidity	NTU	0.8	0.3	0.3	0.1	6784359

MICROBIOLOGY (WATER)

Maxxam ID		GH2230	GH2231	GH2232		
Sampling Date		2013/05/02 08:00	2013/05/02 07:15	2013/05/02 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	<1	1	6791312
Total Coliforms	MPN/100mL	730	11	99	1	6791312

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B336639
Report Date: 2013/05/16

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GI7517	GI7518	GI7519		
Sampling Date		2013/05/08 14:00	2013/05/09 07:30	2013/05/09 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	6804217
Alkalinity (Total as CaCO3)	mg/L	2.4	7.9	4.1	0.5	6800803
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	6800803
Bicarbonate (HCO3)	mg/L	2.9	9.7	5.0	0.5	6800803
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	6800803
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	6800803
MISCELLANEOUS						
Apparent Colour	Col. Unit	19	15	16	5	6808423
True Colour	Col. Unit	11	13	13	5	6808418
Physical Properties						
pH	pH Units	6.4	7.0	6.7		6800783
Physical Properties						
Turbidity	NTU	0.7	0.3	0.3	0.1	6808444

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		GI7517	GI7518	GI7519		
Sampling Date		2013/05/08 14:00	2013/05/09 07:30	2013/05/09 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1060	3500	1650	50	6810432

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B339065
Report Date: 2013/05/24

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GK1314	GK1319	GK1321		
Sampling Date		2013/05/16 07:30	2013/05/16 06:30	2013/05/16 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.7	84.0	82.1	N/A	6820916
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.083	0.076	0.086	0.010	6835179
Dissolved Organic Carbon (C)	mg/L	2.57	3.13	2.61	0.50	6823353
Total Organic Carbon (C)	mg/L	3.05	3.15	2.76	0.50	6823349
MISCELLANEOUS						
Apparent Colour	Col. Unit	16	14	15	5	6824087
True Colour	Col. Unit	12	11	13	5	6824080
Physical Properties						
Total Dissolved Solids	mg/L	19	28	31	10	6827859
Turbidity	NTU	0.8	0.3	0.4	0.1	6823911

MICROBIOLOGY (WATER)

Maxxam ID		GK1314	GK1319	GK1321		
Sampling Date		2013/05/16 07:30	2013/05/16 06:30	2013/05/16 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	3	1	6827947
Total Coliforms	MPN/100mL	870	66	550	1	6827947

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B341263
Report Date: 2013/05/28

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GL5343	GL5344	GL5345		
Sampling Date		2013/05/22 11:40	2013/05/22 12:40	2013/05/23 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	14	13	13	5	6841167
True Colour	Col. Unit	10	10	10	5	6841162
Physical Properties						
Turbidity	NTU	0.8	0.5	0.5	0.1	6841171

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		GL5343	GL5344	GL5345		
Sampling Date		2013/05/22 11:40	2013/05/22 12:40	2013/05/23 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	6846310
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	95	95	93		6846310

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B343637
Report Date: 2013/06/04

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GN1426	GN1427	GN1428		
Sampling Date		2013/05/30 07:30	2013/05/30 08:20	2013/05/30 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	81.5	84.2	70.4	N/A	6859497
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.089	0.075	0.152	0.010	6865444
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	5	13	5	6860587
True Colour	Col. Unit	5	5	13	5	6860584
Physical Properties						
Turbidity	NTU	0.7	0.6	0.8	0.1	6860251

MICROBIOLOGY (WATER)

Maxxam ID		GN1426	GN1427	GN1428		
Sampling Date		2013/05/30 07:30	2013/05/30 08:20	2013/05/30 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	1	3	1	6861538
Total Coliforms	MPN/100mL	230	83	520	1	6861538

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B346268
Report Date: 2013/06/13

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GO7960	GO7961	GO7962		
Sampling Date		2013/06/06 05:40	2013/06/06 06:15	2013/06/06 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	6887801
Alkalinity (Total as CaCO3)	mg/L	2.8	9.0	6.3	0.5	6874807
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	6874807
Bicarbonate (HCO3)	mg/L	3.4	11.0	7.7	0.5	6874807
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	6874807
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	6874807
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	13	11	5	6887056
True Colour	Col. Unit	7	10	10	5	6876396
Physical Properties						
pH	pH Units	6.2	7.0	6.9		6874805
Physical Properties						
Turbidity	NTU	0.6	0.3	0.1	0.1	6880681

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		GO7960	GO7961	GO7962		
Sampling Date		2013/06/06 05:40	2013/06/06 06:15	2013/06/06 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1150	3590	2040	50	6889219

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B348848
Report Date: 2013/06/18

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GQ5640	GQ5641	GQ5642		
Sampling Date		2013/06/13 07:00	2013/06/13 08:50	2013/06/13 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.1	87.8	82.2	N/A	6900677
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.086	0.057	0.085	0.010	6906463
Dissolved Organic Carbon (C)	mg/L	2.58	2.39	2.15	0.50	6906051
Total Organic Carbon (C)	mg/L	2.62	2.54	2.35	0.50	6906048
MISCELLANEOUS						
Apparent Colour	Col. Unit	15	5	15	5	6901500
True Colour	Col. Unit	10	5	10	5	6897198
Physical Properties						
Total Dissolved Solids	mg/L	19	42	37	10	6897200
Turbidity	NTU	0.6	0.3	0.3	0.1	6897202

MICROBIOLOGY (WATER)

Maxxam ID		GQ5640	GQ5641	GQ5642		
Sampling Date		2013/06/13 07:00	2013/06/13 08:50	2013/06/13 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	<1	7	1	6910845
Total Coliforms	MPN/100mL	550	52	770	1	6910845

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B351535
Report Date: 2013/06/27

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GS1922	GS1923	GS1924		
Sampling Date		2013/06/20 10:42	2013/06/20 08:00	2013/06/20 08:05		
	UNITS	**HOLLAND LAKE**	**STOCKING LAKE**	**CHICKEN LADDER**	**RDL**	**QC Batch**
MISCELLANEOUS						
Apparent Colour	Col. Unit	15	10	10	5	6926126
True Colour	Col. Unit	10	10	10	5	6920481
Physical Properties						
Turbidity	NTU	0.8	0.4	0.2	0.1	6926358

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B356187
Report Date: 2013/07/11

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GV0080	GV0081	GV0082		
Sampling Date		2013/07/03 14:30	2013/07/04 08:00	2013/07/04 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	6956714
Alkalinity (Total as CaCO3)	mg/L	2.8	9.3	5.8	0.5	6955483
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	6955483
Bicarbonate (HCO3)	mg/L	3.4	11.3	7.1	0.5	6955483
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	6955483
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	6955483
MISCELLANEOUS						
Apparent Colour	Col. Unit	15	5	10	5	6955598
True Colour	Col. Unit	10	5	10	5	6954299
Physical Properties						
pH	pH Units	6.2	7.0	6.8		6951992
Physical Properties						
Turbidity	NTU	0.8	0.4	1.0	0.1	6959358

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		GV0080	GV0081	GV0082		
Sampling Date		2013/07/03 14:30	2013/07/04 08:00	2013/07/04 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1200	3590	2140	50	6965678

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B354089
Report Date: 2013/07/15

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GT6226	GT6227	GT6228		
Sampling Date		2013/06/27 06:10	2013/06/27 09:10	2013/06/27 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	78.0	83.9	72.6	N/A	6942146
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.108	0.076	0.139	0.010	6946383
MISCELLANEOUS						
Apparent Colour	Col. Unit	15	10	25	5	6942617
True Colour	Col. Unit	15	10	20	5	6940996
Physical Properties						
Turbidity	NTU	0.7	0.5	0.4	0.1	6943881

VOLATILE ORGANICS BY GC-MS (WATER)

Maxxam ID		GT6226	GT6227	GT6228		
Sampling Date		2013/06/27 06:10	2013/06/27 09:10	2013/06/27 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Volatiles						
Bromodichloromethane	ug/L	5.3	4.8	5.0	3.0	6955599
Bromoform	ug/L	<3.0	<3.0	<3.0	3.0	6955599
Chlorodibromomethane	ug/L	<3.0	<3.0	<3.0	3.0	6955599
Chloroform	ug/L	330	310	500	3.0	6955599
THM Formation Potential	ug/L	350	320	520	3.0	6955599

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B358726
Report Date: 2013/07/16

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GW5258	GW5259	GW5260		
Sampling Date		2013/07/11 05:40	2013/07/11 07:00	2013/07/11 07:40		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	81.2	84.7	83.2	N/A	6976714
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.091	0.072	0.080	0.010	6983809
Dissolved Organic Carbon (C)	mg/L	3.10	3.01	1.92	0.50	6982126
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	10	12	5	6977600
True Colour	Col. Unit	5	5	8	5	6977598
Physical Properties						
Total Dissolved Solids	mg/L	13	25	30	10	6982307
Turbidity	NTU	0.7	1.0	0.2	0.1	6977854

MICROBIOLOGY (WATER)

Maxxam ID		GW5258	GW5259	GW5260		
Sampling Date		2013/07/11 05:40	2013/07/11 07:00	2013/07/11 07:40		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	2	4	1	6982128
Total Coliforms	MPN/100mL	550	190	1600	1	6982128

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B361462
Report Date: 2013/07/23

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GY3782	GY3783	GY3784		
Sampling Date		2013/07/17 14:40	2013/07/18 07:50	2013/07/18 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	10	10	5	7000217
True Colour	Col. Unit	5	5	10	5	7000216
Physical Properties						
Turbidity	NTU	1.0	0.6	0.3	0.1	7005266

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		GY3782	GY3783	GY3784		
Sampling Date		2013/07/17 14:40	2013/07/18 07:50	2013/07/18 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	7011625
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	99	97	97		7011625

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B364075
Report Date: 2013/07/29

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HA0657	HA0658		
Sampling Date		2013/07/25 06:40	2013/07/25 08:25		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS					
Transmittance at 254nm	%T/cm	84.3	85.2	N/A	7023724
Misc. Inorganics					
UV absorbance (254nm)	AU/cm	0.074	0.069	0.010	7029588
MISCELLANEOUS					
Apparent Colour	Col. Unit	12	12	5	7024782
True Colour	Col. Unit	10	10	5	7021285
Physical Properties					
Turbidity	NTU	0.4	0.3	0.1	7020335

MICROBIOLOGY (WATER)

Maxxam ID		HA0657	HA0658		
Sampling Date		2013/07/25 06:40	2013/07/25 08:25		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.					
E. coli	MPN/100mL	<1	6	1	7026556
Total Coliforms	MPN/100mL	330	1700	1	7026556

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B366728
Report Date: 2013/08/09

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HB7998	HB7999	HB8000		
Sampling Date		2013/08/01 07:50	2013/08/01 08:00	2013/08/01 08:40		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7043481
Alkalinity (Total as CaCO3)	mg/L	3.3	9.0	5.0	0.5	7039625
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7039625
Bicarbonate (HCO3)	mg/L	4.1	11.0	6.1	0.5	7039625
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7039625
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7039625
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	13	10	5	7046930
True Colour	Col. Unit	10	10	5	5	7046928
Physical Properties						
pH	pH Units	6.5	7.1	6.8		7039622
Physical Properties						
Turbidity	NTU	1.0	0.5	0.3	0.1	7047419

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		HB7998		HB7999	HB8000		
Sampling Date		2013/08/01 07:50		2013/08/01 08:00	2013/08/01 08:40		
	UNITS	HOLLAND LAKE	QC Batch	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS							
Total Calcium (Ca)	ug/L	1260	7055047	4070	1980	50	7054231

RDL = Reportable Detection Limit

Maxxam

Success Through Science®

Maxxam Job #: B369114
Report Date: 2013/08/13

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HD2444	HD2445	HD2446		
Sampling Date		2013/08/08 07:35	2013/08/08 07:25	2013/08/08 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.2	85.8	86.5	N/A	7063538
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.085	0.066	0.063	0.010	7069990
Dissolved Organic Carbon (C)	mg/L	2.48	2.33	1.90	0.50	7069151
MISCELLANEOUS						
Apparent Colour	Col. Unit	13	13	11	5	7063639
True Colour	Col. Unit	5	10	8	5	7062166
Physical Properties						
Total Dissolved Solids	mg/L	20	27	20	10	7066254
Turbidity	NTU	0.7	0.4	0.9	0.1	7062189

MICROBIOLOGY (WATER)

Maxxam ID		HD2444	HD2445	HD2446		
Sampling Date		2013/08/08 07:35	2013/08/08 07:25	2013/08/08 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	1	1	<1	1	7066107
Total Coliforms	MPN/100mL	690	730	1200	1	7066107

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success through Science®

Maxxam Job #: B371943
Report Date: 2013/08/20

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HF0370	HF0371	HF0372		
Sampling Date		2013/08/14 14:05	2013/08/15 08:15	2013/08/15 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Total Organic Carbon (C)	mg/L	2.81	2.77	2.60	0.50	7090407
MISCELLANEOUS						
Apparent Colour	Col. Unit	13	13	15	5	7086274
True Colour	Col. Unit	10	5	15	5	7086254
Physical Properties						
Turbidity	NTU	0.3	0.3	0.1	0.1	7086355

RDL = Reportable Detection Limit

Maxxam Job #: B374659
Report Date: 2013/08/26

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HH0703	HH0704	HH0705		
Sampling Date		2013/08/22 06:40	2013/08/22 08:10	2013/08/22 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.4	85.7	87.5	N/A	7110589
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.084	0.067	0.058	0.010	7114375
MISCELLANEOUS						
Apparent Colour	Col. Unit	14	13	10	5	7110656
True Colour	Col. Unit	12	10	7	5	7109832
Physical Properties						
Turbidity	NTU	0.6	0.3	0.1	0.1	7107420

MICROBIOLOGY (WATER)

Maxxam ID		HH0703	HH0704	HH0705		
Sampling Date		2013/08/22 06:40	2013/08/22 08:10	2013/08/22 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	2	1	1	7115138
Total Coliforms	MPN/100mL	340	980	1700	1	7115138

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam Job #: B377406
Report Date: 2013/09/09

TOWN OF LADYSMITH

Your P.O.#: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HI9813	HI9814	HI9815		
Sampling Date		2013/08/28 14:00	2013/08/29 08:10	2013/08/29 07:55		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7130091
Alkalinity (Total as CaCO3)	mg/L	3.4	8.9	5.1	0.5	7126144
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7126144
Bicarbonate (HCO3)	mg/L	4.1	10.8	6.2	0.5	7126144
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7126144
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7126144
MISCELLANEOUS						
Apparent Colour	Col. Unit	17	16	16	5	7132006
True Colour	Col. Unit	12	10	11	5	7132004
Physical Properties						
pH	pH Units	6.6	6.9	6.6		7126139
Physical Properties						
Turbidity	NTU	0.6	0.6	0.2	0.1	7132050

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		HI9813	HI9814	HI9815		
Sampling Date		2013/08/28 14:00	2013/08/29 08:10	2013/08/29 07:55		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1260	4240	2160	50	7146508

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B379813
Report Date: 2013/09/12

TOWN OF LADYSMITH
Client Project #: YEARLY WATER

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HK4793	HK4794	HK4795		
Sampling Date		2013/09/05 06:30	2013/09/05 08:20	2013/09/05 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	83.2	85.1	87.1	N/A	7150021
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.080	0.070	0.060	0.010	7163557
Dissolved Organic Carbon (C)	mg/L	2.66	2.74	1.98	0.50	7153800
Total Organic Carbon (C)	mg/L	3.04	2.81	2.07	0.50	7153789
MISCELLANEOUS						
Apparent Colour	Col. Unit	13	14	14	5	7149983
True Colour	Col. Unit	9	10	13	5	7149889
Physical Properties						
Total Dissolved Solids	mg/L	24	30	23	10	7156246
Turbidity	NTU	0.7	0.5	0.1	0.1	7149885

MICROBIOLOGY (WATER)

Maxxam ID		HK4793	HK4794	HK4795		
Sampling Date		2013/09/05 06:30	2013/09/05 08:20	2013/09/05 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	2	2	4	1	7151277
Total Coliforms	MPN/100mL	730	2400	1300	1	7151277

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B382543
Report Date: 2013/09/27

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HM3822	HM3823	HM3824		
Sampling Date		2013/09/12 07:00	2013/09/11 13:20	2013/09/12 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	19	17	12	5	7169458
True Colour	Col. Unit	13	11	10	5	7169457
Physical Properties						
Turbidity	NTU	0.9	0.6	0.2	0.1	7164773

VOLATILE ORGANICS BY GC-MS (WATER)

Maxxam ID		HM3822	HM3823	HM3824		
Sampling Date		2013/09/12 07:00	2013/09/11 13:20	2013/09/12 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Volatiles						
Bromodichloromethane	ug/L	5.5	4.1	5.3	3.0	7176385
Bromoform	ug/L	<3.0	<3.0	<3.0	3.0	7176385
Chlorodibromomethane	ug/L	<3.0	<3.0	<3.0	3.0	7176385
Chloroform	ug/L	260	290	180	3.0	7176385
THM Formation Potential	ug/L	270	300	190	3.0	7176385

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B385267
Report Date: 2013/09/26

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HO4275	HO4276		
Sampling Date		2013/09/19 08:10	2013/09/19 08:05		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS					
Transmittance at 254nm	%T/cm	86.2	88.1	N/A	7181574
Misc. Inorganics					
UV absorbance (254nm)	AU/cm	0.064	0.055	0.010	7192948
MISCELLANEOUS					
Apparent Colour	Col. Unit	13	14	5	7181628
True Colour	Col. Unit	10	10	5	7181621
Physical Properties					
Turbidity	NTU	0.6	0.3	0.1	7183258

MICROBIOLOGY (WATER)

Maxxam ID		HO4275	HO4276		
Sampling Date		2013/09/19 08:10	2013/09/19 08:05		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.					
E. coli	MPN/100mL	1	2	1	7182907
Total Coliforms	MPN/100mL	2000	1700	1	7182907

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B388126
Report Date: 2013/10/03

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HQ3278	HQ3279	HQ3280		
Sampling Date		2013/09/25 14:02	2013/09/26 08:05	2013/09/26 08:30		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7194706
Alkalinity (Total as CaCO3)	mg/L	3.0	9.1	4.7	0.5	7191687
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7191687
Bicarbonate (HCO3)	mg/L	3.6	11.1	5.7	0.5	7191687
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7191687
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7191687
MISCELLANEOUS						
Apparent Colour	Col. Unit	16	15	13	5	7196451
True Colour	Col. Unit	11	10	9	5	7196447
Physical Properties						
pH	pH Units	6.6	7.1	6.8		7191685
Physical Properties						
Turbidity	NTU	0.6	0.4	0.3	0.1	7196742

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		HQ3278		HQ3279	HQ3280		
Sampling Date		2013/09/25 14:02		2013/09/26 08:05	2013/09/26 08:30		
	UNITS	HOLLAND LAKE	QC Batch	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS							
Total Calcium (Ca)	ug/L	1420	7201765	4560	2030	50	7199479

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B390814
Report Date: 2013/10/09

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HS1964	HS1965		HS1966		
Sampling Date		2013/10/03 06:40	2013/10/03 09:00		2013/10/03 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	QC Batch	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS							
Transmittance at 254nm	%T/cm	81.7	86.7	7207541	64.3	N/A	7207541
Misc. Inorganics							
UV absorbance (254nm)	AU/cm	0.088	0.062	7216007	0.192	0.010	7216007
Dissolved Organic Carbon (C)	mg/L	2.98	2.47	7211615	5.34	0.50	7211615
Total Organic Carbon (C)	mg/L	4.01	3.92	7211611	5.65	0.50	7211611
MISCELLANEOUS							
Apparent Colour	Col. Unit	15	15	7209323	25	5	7209323
True Colour	Col. Unit	9	10	7208669	20	5	7208669
Physical Properties							
Total Dissolved Solids	mg/L	22	50	7207910	62	10	7207910
Turbidity	NTU	0.7	0.5	7205775	0.6	0.1	7208605

MICROBIOLOGY (WATER)

Maxxam ID		HS1964	HS1965	HS1966		
Sampling Date		2013/10/03 06:40	2013/10/03 09:00	2013/10/03 08:10		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	15	3	12	1	7209317
Total Coliforms	MPN/100mL	310	730	1300	1	7209317

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B393527
Report Date: 2013/10/16

TOWN OF LADYSMITH
Client Project #: YEARLY H2O

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HU1542	HU1543	HU1544		
Sampling Date		2013/10/09	2013/10/10 08:45	2013/10/10 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	12	5	13	5	7222847
True Colour	Col. Unit	10	5	13	5	7222845
Physical Properties						
Turbidity	NTU	0.7	0.3	0.4	0.1	7222835

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		HU1542	HU1543	HU1544		
Sampling Date		2013/10/09	2013/10/10 08:45	2013/10/10 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	7227502
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	102	102	101		7227502

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B395790
Report Date: 2013/10/24

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HV6579	HV6580	HV6581		
Sampling Date		2013/10/17 07:15	2013/10/17 08:50	2013/10/17 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.6	86.5	80.7	N/A	7234622
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.083	0.063	0.093	0.010	7242216
MISCELLANEOUS						
Apparent Colour	Col. Unit	15	13	12	5	7235741
True Colour	Col. Unit	10	8	12	5	7235738
Physical Properties						
Turbidity	NTU	1.2	0.5	0.8	0.1	7235735

MICROBIOLOGY (WATER)

Maxxam ID		HV6579	HV6580	HV6581		
Sampling Date		2013/10/17 07:15	2013/10/17 08:50	2013/10/17 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	1	1	1	1	7238906
Total Coliforms	MPN/100mL	110	140	610	1	7238906

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam Job #: B398432
Report Date: 2013/10/31

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HX4316	HX4317	HX4318		
Sampling Date		2013/10/23 13:30	2013/10/24 09:00	2013/10/24 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7250776
Alkalinity (Total as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Bicarbonate (HCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	10	16	5	7250860
True Colour	Col. Unit	10	10	15	5	7247816
Physical Properties						
pH	pH Units	3.3	3.8	4.0		7252184
Physical Properties						
Turbidity	NTU	0.7	0.2	0.3	0.1	7247795

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		HX4316	HX4317		HX4318		
Sampling Date		2013/10/23 13:30	2013/10/24 09:00		2013/10/24 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	QC Batch	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS							
Total Calcium (Ca)	ug/L	1170	4730	7253917	2560	50	7255165

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B398432
Report Date: 2013/11/01

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HX4316	HX4317	HX4318		
Sampling Date		2013/10/23 13:30	2013/10/24 09:00	2013/10/24 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7250776
Alkalinity (Total as CaCO3)	mg/L	2.8	10.6	6.6	0.5	7252187
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Bicarbonate (HCO3)	mg/L	3.4	12.9	8.0	0.5	7252187
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7252187
MISCELLANEOUS						
Apparent Colour	Col. Unit	10	10	16	5	7250860
True Colour	Col. Unit	10	10	15	5	7247816
Physical Properties						
pH	pH Units	6.5	7.1	6.9		7252184
Physical Properties						
Turbidity	NTU	0.7	0.2	0.3	0.1	7247795

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		HX4316	HX4317		HX4318		
Sampling Date		2013/10/23 13:30	2013/10/24 09:00		2013/10/24 08:00		
	UNITS	HOLLAND LAKE	STOCKING LAKE	QC Batch	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS							
Total Calcium (Ca)	ug/L	1170	4730	7253917	2560	50	7255165

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B3A1064
Report Date: 2013/11/07

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HZ3474	HZ3475	HZ3476		
Sampling Date		2013/10/30 13:40	2013/10/31 08:30	2013/10/31 07:55		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	84.1	88.5	85.7	N/A	7258907
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.075	0.053	0.067	0.010	7263157
Dissolved Organic Carbon (C)	mg/L	2.31(1)	2.27(1)	2.00(1)	0.50	7262480
Total Organic Carbon (C)	mg/L	3.48	1.93	1.88	0.50	7266992
MISCELLANEOUS						
Apparent Colour	Col. Unit	11	10	14	5	7259903
True Colour	Col. Unit	5	7	11	5	7259904
Physical Properties						
Total Dissolved Solids	mg/L	23	27	23	10	7261571
Turbidity	NTU	0.9	0.3	0.2	0.1	7259489

MICROBIOLOGY (WATER)

Maxxam ID		HZ3475	HZ3476		
Sampling Date		2013/10/31 08:30	2013/10/31 07:55		
	UNITS	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.					
E. coli	MPN/100mL	<1	21	1	7259245
Total Coliforms	MPN/100mL	28	690	1	7259245

N/A = Not Applicable
RDL = Reportable Detection Limit
(1) - Sample analysed past recommended hold time.

Maxxam

Success Through Science

Maxxam Job #: B3A3807
Report Date: 2013/11/12

TOWN OF LADYSMITH
Client Project #: YEARLY WATER TESTING

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		IA9680	IA9681	IA9682		
Sampling Date		2013/11/07 05:50	2013/11/07 08:10	2013/11/07 08:40		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
MISCELLANEOUS						
Apparent Colour	Col. Unit	11	8	40	5	7271115
True Colour	Col. Unit	9	6	25	5	7271112
Physical Properties						
Turbidity	NTU	0.8	0.5	2.4	0.1	7271109

MICROBIOLOGY (WATER)

Maxxam ID		IA9680		
Sampling Date		2013/11/07 05:50		
	UNITS	HOLLAND LAKE	RDL	QC Batch
Microbiological Param.				
E. coli	MPN/100mL	1	1	7271188
Total Coliforms	MPN/100mL	210	1	7271188

RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B3A6044
Report Date: 2013/11/20

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		IC2750	IC2751	IC2752		
Sampling Date		2013/11/13 14:00	2013/11/13 11:00	2013/11/13 13:30		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	83.1	84.8	68.7	N/A	7282257
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.081	0.072	0.163	0.010	7286926
MISCELLANEOUS						
Apparent Colour	Col. Unit	11	10	30	5	7282361
True Colour	Col. Unit	12	10	30	5	7282360
Physical Properties						
Turbidity	NTU	0.7	0.5	0.2	0.1	7282286

MICROBIOLOGY (WATER)

Maxxam ID		IC2750	IC2751	IC2752		
Sampling Date		2013/11/13 14:00	2013/11/13 11:00	2013/11/13 13:30		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
E. coli	MPN/100mL	<1	1	3	1	7283107
Total Coliforms	MPN/100mL	360	59	520	1	7283107

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B3A8393
Report Date: 2013/11/26

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		ID7322	ID7323	ID7324		
Sampling Date		2013/11/21 08:40	2013/11/21 08:05	2013/11/21 08:40		
	UNITS	**HOLLAND LAKE**	**STOCKING LAKE**	**CHICKEN LADDER**	**RDL**	**QC Batch**
MISCELLANEOUS						
Apparent Colour	Col. Unit	13	11	15	5	7294081
True Colour	Col. Unit	11	9	14	5	7294080
Physical Properties						
Turbidity	NTU	0.9	0.5	0.6	0.1	7291217

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B3B0562
Report Date: 2013/12/05

TOWN OF LADYSMITH
Client Project #: RAW WATER NOV 28/13

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		IF0525	IF0526	IF0527		
Sampling Date		2013/11/28 06:10	2013/11/28 09:00	2013/11/28 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.5	88.7	80.6	N/A	7302532
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.083	0.052	0.094	0.010	7308209
Dissolved Organic Carbon (C)	mg/L	2.73(1)	1.65(1)	2.58(1)	0.50	7307626
Total Organic Carbon (C)	mg/L	3.06	2.27	2.79	0.50	7307623
MISCELLANEOUS						
Apparent Colour	Col. Unit	11	10	13	5	7305325
True Colour	Col. Unit	10	10	12	5	7301373
Physical Properties						
Total Dissolved Solids	mg/L	13	27	29	10	7302224
Turbidity	NTU	0.7	0.2	<0.1	0.1	7306766

MICROBIOLOGY (WATER)

Maxxam ID		IF0525	IF0526	IF0527		
Sampling Date		2013/11/28 06:10	2013/11/28 09:00	2013/11/28 08:05		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Microbiological Param.						
Total Coliforms	CFU/100mL	73	15	110	1	7302254
E. coli	CFU/100mL	1	<1	<1	1	7302254

N/A = Not Applicable
RDL = Reportable Detection Limit
(1) - Sample analysed past recommended hold time.

Maxxam

Maxxam Job #: B3B2812
Report Date: 2013/12/12

TOWN OF LADYSMITH
Client Project #: RAW WATER YEARLY SAMPLES DEC 5

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		IG2256	IG2257	IG2258		
Sampling Date		2013/12/04 13:45	2013/12/05 08:15	2013/12/05 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Misc. Inorganics						
Dissolved Bromide (Br)	mg/L	<0.010	<0.010	<0.010	0.010	7309235
Alkalinity (Total as CaCO3)	mg/L	3.0	9.4	6.2	0.5	7314158
Alkalinity (PP as CaCO3)	mg/L	<0.5	<0.5	<0.5	0.5	7314158
Bicarbonate (HCO3)	mg/L	3.6	11.5	7.6	0.5	7314158
Carbonate (CO3)	mg/L	<0.5	<0.5	<0.5	0.5	7314158
Hydroxide (OH)	mg/L	<0.5	<0.5	<0.5	0.5	7314158
MISCELLANEOUS						
Apparent Colour	Col. Unit	14	10	12	5	7311519
True Colour	Col. Unit	10	5	10	5	7309759
Physical Properties						
pH	pH	6.5	7.0	6.8		7314135
Physical Properties						
Turbidity	NTU	0.7	0.5	0.2	0.1	7309536

ELEMENTS BY ATOMIC SPECTROSCOPY (WATER)

Maxxam ID		IG2256	IG2257	IG2258		
Sampling Date		2013/12/04 13:45	2013/12/05 08:15	2013/12/05 08:35		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Total Metals by ICPMS						
Total Calcium (Ca)	ug/L	1400	1330	2730	50	7314926

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B3B4982
Report Date: 2013/12/19

TOWN OF LADYSMITH
Client Project #: RAW WATER DEC 12/13

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		IH4295	IH4296	IH4297		
Sampling Date		2013/12/12 08:20	2013/12/12 09:00	2013/12/12 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
CONVENTIONALS						
Transmittance at 254nm	%T/cm	82.8	86.7	85.2	N/A	7320969
Misc. Inorganics						
UV absorbance (254nm)	AU/cm	0.082	0.062	0.069	0.010	7325507
MISCELLANEOUS						
Apparent Colour	Col. Unit	12	13	12	5	7320978
True Colour	Col. Unit	10	10	10	5	7319687
Physical Properties						
Turbidity	NTU	0.2	0.4	0.7	0.1	7319239

TOTAL PETROLEUM HYDROCARBONS (WATER)

Maxxam ID		IH4295	IH4296	IH4297		
Sampling Date		2013/12/12 08:20	2013/12/12 09:00	2013/12/12 08:15		
	UNITS	HOLLAND LAKE	STOCKING LAKE	CHICKEN LADDER	RDL	QC Batch
Hydrocarbons						
Total Extractables C10 to C30	mg/L	<0.20	<0.20	<0.20	0.20	7325246
Surrogate Recovery (%)						
O-TERPHENYL (sur.)	%	113	113	111		7325246

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success through Science

Maxxam Job #: B3B7017
Report Date: 2013/12/24

TOWN OF LADYSMITH
Client Project #: YEARLEY WATER DEC 19/13

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		II5432	II5433	II5434		
Sampling Date		2013/12/18 13:20	2013/12/19 08:25	2013/12/19 08:50		
	UNITS	**HOLLAND LAKE**	**STOCKING LAKE**	**CHICKEN LADDER**	**RDL**	**QC Batch**
MISCELLANEOUS						
Apparent Colour	Col. Unit	17	12	15	5	7330896
True Colour	Col. Unit	12	10	10	5	7330889
Physical Properties						
Total Dissolved Solids	mg/L	14	13	21	10	7330778
Turbidity	NTU	0.7	0.5	0.6	0.1	7330634

RDL = Reportable Detection Limit

Maxxam

Maxxam Job #: B318285
Report Date: 2013/03/21

TOWN OF LADYSMITH
Client Project #: MARCH 2013 HAA & THM

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		FU8573	FU8574		
Sampling Date		2013/03/07 08:25	2013/03/07 08:00		
	UNITS	558 HOOPER PLACE	1280 ROCKY CREEK	RDL	QC Batch
Parameter					
Subcontract Parameter	N/A	ATTACHED	ATTACHED	N/A	6672742

TRIHALOMETHANES (THM) IN WATER

Maxxam ID		FU8573	FU8574		
Sampling Date		2013/03/07 08:25	2013/03/07 08:00		
	UNITS	558 HOOPER PLACE	1280 ROCKY CREEK	RDL	QC Batch
Volatiles					
Chloroform	ug/L	76	68	1.0	6639239
Chlorodibromomethane	ug/L	<1.0	<1.0	1.0	6639239
Bromodichloromethane	ug/L	2.2	2.0	1.0	6639239
Bromoform	ug/L	<1.0	<1.0	1.0	6639239
Surrogate Recovery (%)					
1,4-Difluorobenzene (sur.)	%	99	101		6639239
4-BROMOFLUOROBENZENE (sur.)	%	88	95		6639239
D4-1,2-DICHLOROETHANE (sur.)	%	110	113		6639239

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B349990
Report Date: 2013/07/03

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		GR3129	GR3130		
Sampling Date		2013/06/17 08:30	2013/06/17 08:00		
	UNITS	**558 HOOPER PLACE**	**1280 ROCKY CREEK**	**RDL**	**QC Batch**
Parameter					
Subcontract Parameter	N/A	ATTACHED	ATTACHED	N/A	6947762

TRIHALOMETHANES (THM) IN WATER

Maxxam ID		GR3129	GR3130		
Sampling Date		2013/06/17 08:30	2013/06/17 08:00		
	UNITS	**558 HOOPER PLACE**	**1280 ROCKY CREEK**	**RDL**	**QC Batch**
Volatiles					
Chloroform	ug/L	66	67	1.0	6921462
Chlorodibromomethane	ug/L	<1.0	<1.0	1.0	6921462
Bromodichloromethane	ug/L	1.9	1.8	1.0	6921462
Bromoform	ug/L	<1.0	<1.0	1.0	6921462
Surrogate Recovery (%)					
1.4-Difluorobenzene (sur.)	%	99	104		6921462
4-BROMOFLUOROBENZENE (sur.)	%	81	94		6921462
D4-1,2-DICHLOROETHANE (sur.)	%	82	95		6921462

N/A = Not Applicable
RDL = Reportable Detection Limit

Maxxam

Success Through Science

Maxxam Job #: B384680
Report Date: 2013/10/04

TOWN OF LADYSMITH

Your P.O. #: 10429

RESULTS OF CHEMICAL ANALYSES OF WATER

Maxxam ID		HO0567	HO0568		
Sampling Date		2013/09/18 07:55	2013/09/18 08:10		
	UNITS	558 HOOPER PLACE	1280 ROCKY CREEK	RDL	QC Batch
Parameter					
Subcontract Parameter	N/A	ATTACHED	ATTACHED	N/A	7208003

TRIHALOMETHANES (THM) IN WATER

Maxxam ID		HO0567	HO0568		
Sampling Date		2013/09/18 07:55	2013/09/18 08:10		
	UNITS	558 HOOPER PLACE	1280 ROCKY CREEK	RDL	QC Batch
Volatiles					
Chloroform	ug/L	71	62	1.0	7184675
Chlorodibromomethane	ug/L	<1.0	<1.0	1.0	7184675
Bromodichloromethane	ug/L	3.6	3.2	1.0	7184675
Bromoform	ug/L	<1.0	<1.0	1.0	7184675
Surrogate Recovery (%)					
1,4-Difluorobenzene (sur.)	%	103	104		7184675
4-BROMOFLUOROBENZENE (sur.)	%	89	92		7184675
D4-1,2-DICHLOROETHANE (sur.)	%	104	105		7184675

N/A = Not Applicable
RDL = Reportable Detection Limit